



Illuminating
ENGINEERING SOCIETY

APPROVED METHOD:
**PHOTOMETRIC TESTING OF ROADWAY
AND AREA LIGHTING LUMINAIRES USING
INCANDESCENT FILAMENT OR HIGH
INTENSITY DISCHARGE LAMPS**
AN AMERICAN NATIONAL STANDARD



Note: This 2023 Re-affirmation corrects an editorial error from the 2020 version:

Section 6.5.4.5.5 Original Text:

If the light distribution of the luminaire exhibits bilateral, four-quadrant, or octant symmetry, appropriate averaging as described in **Section 6.5.4.2.2** may be applied.

Section 6.5.4.5.5 Corrected Text:

If the light distribution of the luminaire exhibits bilateral, four-quadrant, or octant symmetry, appropriate averaging as shown in **Figures 6-8, 6-9, and 6-10** may be applied.

ANSI/IES LM-31-20(R2023)

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Publication of this Committee
report has been approved by IES.
Suggestions for revision should
be directed to IES.

**Prepared by
The IES Testing Procedures Committee**



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CONTENTS

Foreword	1
1.0 Introduction and Scope	1
1.1 Introduction	1
1.2 Scope	1
2.0 Normative References	1
2.1 ANSI/IES LM-28-20	1
2.2 ANSI/IES LM-45-20	1
2.3 ANSI/IES LM-51-20	1
2.4 ANSI/IES LM-54-20	1
2.5 ANSI/IES LS-1-20	1
3.0 Nomenclature and Definitions	1
3.1 auxiliary Equipment	2
3.2 area lighting	2
3.3 pre-burning	2
3.4 test distance	2
3.5 units	2
4.0 Physical and Environmental Test Conditions	2
4.1 General	2
4.2 Temperature	2
4.3 Airflow	2
4.4 Stray Light	2
4.5 Vibration	3
5.0 Electrical Test Conditions	3
5.1 Power Supply Requirements	3
5.1.1 Primary Source Voltage	3
5.1.2 Power Supply Wave Shape	3
5.1.3 Power Supply Regulation	3
5.2 Reference Circuit Requirements	4
5.3 Measurement Instrument Requirements	4

6.0	Testing Procedure Requirements	4
6.1	General	4
6.1.1	Goniophotometer	4
6.2	Test Distance	5
6.3	Selection and Preparation of Luminaire	6
6.3.1	Luminaire, Lamp, and Ballast Selection	6
6.3.2	Cleaning of Luminaire Optical Parts	6
6.4	Test Lamps	6
6.4.1	General	6
6.4.2	Lamp Seasoning	6
6.4.3	Test Lamp Operation	6
6.5	Test Methodologies	7
6.5.1	General	7
6.5.2	Bare Lamp Measurement	8
6.5.3	Lamp Photometric Calibration Process	8
6.5.4	Luminaire Photometry	12
6.5.5	Photometric Data Processing	17
7.0	Test Report	19
7.1	General	19
7.2	Test Description	19
Annex A	Zonal Constants for Conversion	20
Annex B	Procedure for Computing Iso-illuminance Curves	21
Annex C	Computing Utilization Efficiency	21
Annex D	Sample Test Report	25
Annex E	Table of Constants for Horizontal Illuminance	33
Additional Reading		34
Informative References		35

Foreword

This approved testing methods guide is a revision of LM-31-1995. Significant changes have been made in the revision. In particular the structure of the revised LM follows the IES Testing Procedure guidelines for organizing an LM that uses a structure more suitable to a standard. Further, there has been an attempt to harmonize the wording in this revision with revised versions of *ANSI/IES LM-10-20/R23, Approved Method: Photometric Testing of Roadway and Area Lighting Fluorescent Luminaires*; *ANSI/IES LM-35-20/R23, Approved Method: Photometric Testing of Floodlights Using High Intensity Discharge or Incandescent Filament Lamps*; and *ANSI/IES LM-46-20/R23, Approved Method: Photometric Testing of Indoor Luminaires Using HID or Incandescent Filament Lamps*.

1.0 Introduction and Scope

1.1 Introduction

Exterior illumination provided by luminaires generally falls into the standard Type I, II, III, IV, V and VS classifications. Distribution patterns are typically bisymmetric, quadrilaterally symmetric, or completely symmetric. Illumination provided by luminaires such as outdoor wall packs, bollards, and unique cases of general area lighting may not be designed to fit into the standard classifications and may have completely asymmetric distribution patterns.

The test procedures for luminaires using incandescent filament or high intensity discharge (HID) lamps described herein can utilize either absolute or relative photometry methods. In both cases, general conditions of suitability such as test distance should be assessed for adequacy to ensure the validity of the test results.

1.2 Scope

This Lighting Measurement (LM) guide defines adequate and uniform methods for measuring and reporting the photometric characteristics of roadway and area lighting luminaires using incandescent filament or high intensity discharge lamps. It describes characteristics of luminaires and some components, as well as the requirements for

the thermal environment and proper control of the electrical and mechanical systems involved. General test conditions and the testing procedure best suited for achieving accurate and consistent photometric results are defined.

This guide does not cover testing of luminaires using low pressure sodium lamps.

2.0 Normative References

2.1 ANSI/IES LM-28-20

Illuminating Engineering Society. Approved Method: Guide for the Selection, Care, and Use of Electrical Instruments in the Photometric Laboratory. New York: IES; 2020.

2.2 ANSI/IES LM-45-20

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurement of General Service Incandescent Filament Lamps. New York: IES; 2020.

2.3 ANSI/IES LM-51-20

Illuminating Engineering Society. Approved Method: The Electrical and Photometric Measurements of High Intensity Discharge Lamps. New York: IES; 2020.

2.4 ANSI/IES LM-54-20

Illuminating Engineering Society. IES Guide to Lamp Seasoning. New York: IES; 2020.

2.5 ANSI/IES LS-1-20

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: <https://www.ies.org/standards/definitions/>. (Accessed 2019 Sep 18).

3.0 Nomenclature and Definitions

The terms used in this document follow the definitions given in **Normative Reference 2.5**. Additional terms are defined below.

3.1 auxiliary equipment

The auxiliary equipment includes the ballast, capacitor, and the igniter needed for the proper operation of discharge lamps.

3.2 area lighting

In this document, *area lighting* refers to exterior illumination provided by luminaires (such as outdoor wall packs, bollards, and general area lighting luminaires) that may not be designed to fit into the standard Type I, II, III, IV, V, or VS classifications.

3.3 pre-burning

Pre-burning is defined as operating a previously seasoned lamp for a sufficient period of time such that it reaches thermal equilibrium, the mercury locates at the coolest zone of the lamp, and a stable light output is achieved under the test conditions.

3.4 test distance

The *test distance* shall be defined as the distance along the light path (direct and/or reflected) from the center of rotation of the goniophotometer axes as shown in **Figure 6-2 (Section 6.2)**, to the surface of the photodetector.

3.5 units

The units of electrical measurement used in this approved method are the volt, ampere, and watt. The units of photometric measurement are the lumen and the candela. Color is specified using chromaticity via chromaticity coordinates. 1976 (*u'*, *v'*) chromaticity coordinates are preferred, but 1931 (*x*, *y*) chromaticity coordinates may also be used.

4.0 Physical and Environmental Test Conditions

4.1 General

Changes in ambient temperature or air movement have little influence on the electrical and photometric characteristics of incandescent filament and HID lamps. In the case of HID lamps, the discharge tube is encapsulated inside an outer bulb, which provides a

level of thermal and airflow isolation. Thus, the luminous flux remains almost constant over a wide range of ambient temperatures. It is good laboratory practice that the storage and testing of lamps be undertaken in a relatively clean environment. For lamps that operate at a high bulb temperature, contaminants can be etched into glass bulb surfaces during operation. Therefore, lamps should be cleaned before measurement, and operators should use gloves when handling them. It is important to note that some gloves are powdered to avoid self-adhesion. These powdered gloves should be avoided since the powder may leave behind an undesirable residue on the lamp.

4.2 Temperature

Referring to **Figure 4-1**, the ambient temperature of the photometric laboratory within 1 meter of the luminaire in Zone 2 shall be maintained at $25\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$ ($77\text{ }^{\circ}\text{F} \pm 9\text{ }^{\circ}\text{F}$). The ambient temperature shall be measured at the same height and not more than 1.0 meter (3.3 feet) from the luminaire. The ambient temperature in the goniophotometer room, as shown in *Zone 1* shall be maintained at $25\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$ ($77\text{ }^{\circ}\text{F} \pm 9\text{ }^{\circ}\text{F}$). Temperature sensors shall be shielded from direct radiation and convection currents produced by the luminaire.

4.3 Airflow

The incidence of air movement on the surface of an SSL product under test may substantially alter electrical and photometric values. Airflow around the SSL product under test should be such that normal convective airflow induced by the device under test is not affected.

4.4 Stray Light

Stray light should be suppressed in the test environment through the adequate use of low-reflectance surface finishes, shielding and baffling. Methods for stray light suppression include limiting the field of view, reducing reflected light, and using light traps.² The walls, ceiling and floor of the photometric test room should be painted flat black or covered with a diffuse, low-reflectance material such as black felt, black velvet, or black carpeting. It is important to note that the directional response of some materials should also be considered. Opaque low-reflectance baffles with limiting apertures interposed between the photodetector position and the luminaire position that shield the photodetector,

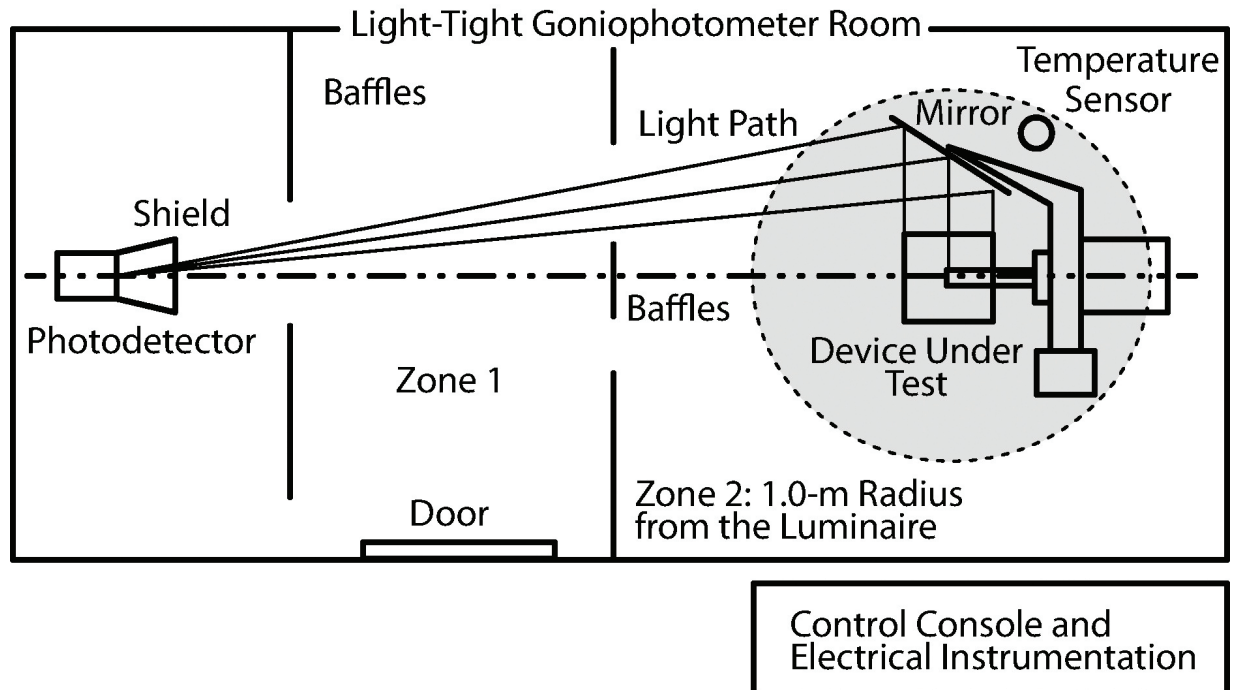


Figure 4-1. Layout of a typical moving-mirror goniophotometer. (© Illuminating Engineering Society)

except in the direction of the test source, should aid in suppressing the stray light reaching the photodetector. For moving-photodetector goniophotometers, the light incident on the photodetector should only be that which is directly transmitted from the luminaire or bare lamp. For moving-mirror goniophotometers, only the light directly from the luminaire or bare lamp reflected via the mirror should be measured. Stray light may be measured by running a complete or partial test with all direct light from the lamp or luminaire shielded from the photodetector. The choice of partial or complete measurement of the stray light depends on the spatial properties of the luminaire's photometric pattern and the expected stray light contribution at each measured vertical and horizontal angle pair. The measured stray light, if any, shall then be subtracted from the bare lamp or luminaire test data, at each vertical angle in each plane measured.

4.5 Vibration

Lamps should not be subjected to excessive vibration or shock during measurement.

5.0 Electrical Test Conditions

5.1 Power Supply Requirements

5.1.1 Primary Source Voltage. In the case of HID lamps, ballasts should be operated at their nominal rated primary voltage and frequency. If the voltage rating is a range, the ballast should be operated at the intended application voltage, and the test voltage shall be recorded in the test report.

5.1.2 Power Supply Wave Shape. The AC power supply, while operating the test lamp(s) or luminaire, shall have a sinusoidal voltage wave shape such that the value of the root-mean-square (RMS) summation of the harmonic components shall not exceed 3% of the fundamental. Caution should be exercised when using constant voltage transformer regulators. This type of system can produce harmonic distortion in excess of 3% when lightly loaded.

5.1.3 Power Supply Regulation. During testing, lamp wattage shall be regulated to stabilize light output. Lamp wattage control is achieved via power supply

regulation. The method used depends on the type of ballast and lamp used for testing. For incandescent lamps, lamp wattage is controlled via current regulation. For HID lamps running on electronic ballasts, the ballast electronics act as a wattage regulator, and only the supply voltage is regulated during testing. For HID lamps running on magnetic ballasts, wattage is regulated via supply voltage regulation.

In all cases, to ensure stable luminous output from the lamp, the regulated parameter, either current or voltage, should be tightly regulated. (See **Section 6.4.3** for regulation details for each lamp type.)

5.2 Reference Circuit Requirements

For HID lamps, ballasts regularly furnished as part of or recommended for use with the luminaire may be used to operate the lamp(s) during the test. As an alternative, American National Standards Institute (ANSI) compliant “reference ballasts” may also be used. Due to the size of the reference ballasts, testing is performed with the reference ballast external to the luminaire.

5.3 Measurement Instrument Requirements

Electrical parameters (e.g., voltage, current, and power) shall be measured with instruments having calibration that is traceable to a national metrological institute (NMI). The instrument measurement tolerance* of alternating current (AC) voltage and current shall be less than or equal to 0.2%. The instrument measurement tolerance of AC power shall be less than or equal to 0.5%. The instrument measurement tolerance of direct current (DC) voltage and current shall be less than or equal to 0.1%.

Instrument measurement ranges should be selected so that the mid-to-maximum portion of the range is used for any specific measurement. Since gaseous discharge lamps may have greatly distorted wave shapes, AC instruments (for voltage, current, and power) shall measure true root-mean-square (RMS) values. Instruments with scales calibrated in RMS values but with deflections or readings

based on average or peak values shall not be used. If analog instrumentation is used, large-scale deflection for the intended measurement range should be used.

The electrical instruments should be chosen according to the guidelines and precautions presented in ANSI/IES LM-28-20 (see **Section 2.1**). It should be noted that some HID electronic ballasts provide a high-frequency lamp current. Care should be taken to ensure that possible conducted or radiated electromagnetic interference (EMI) from this type of ballast does not cause reading errors. High frequency wire runs should be as short as possible and, if necessary, shielded from EMI. It may be necessary to use precautions in order to prevent EMI from reaching voltage monitoring equipment or the photodetector readout device through the power circuits.

6.0 Testing Procedure Requirements

6.1 General

The luminaire shall be operated under conditions as nearly typical of end use as possible, or suitable corrections made. For example, if a luminaire is intended to be mounted on pole at a tilted angle, it shall be mounted at that same angle during testing.

6.1.1 Goniophotometer. A goniophotometer¹ is a partially or fully automated electro-mechanical system that positions a lamp and/or luminaire and measures the directional distribution of light. The goniophotometer and mounting apparatus should be rigid enough to prevent angular deflection of the lamp and/or luminaire and the goniophotometer structure even when there is appreciable off-balance loading from the lamp and/or luminaire and its mounting apparatus. The goniophotometer and associated mounting apparatus should minimize interference with light emitted by the lamp and/or luminaire. Type C goniophotometers shall be used to avoid light output change due to lamp tilting.

6.1.1.1 Photodetectors. The photometer head, or photodetector,² of the goniophotometer should have relative spectral responsivity matched to the $V(\lambda)$ function. The f_1' value of the spectral responsivity shall be less than 3%. For absolute measurements on

* Expanded uncertainty with a 95% confidence interval, normally with a coverage factor $k=2$, as prescribed in *JCGM 100:2008, Evaluation of measurement data - Guide to Expression of Uncertainty in Measurement*, and *ANSI/NCSL Z540-2-1997, U.S. Guide to the Expression of Uncertainty in Measurement*. If a manufacturer's specification does not specify uncertainty this way, then the manufacturer should be contacted for proper conversion.

spectrally structured sources (for example, most HID sources) a systematic spectral correction factor should be considered. To determine such a correction factor, the spectral responsivity of the photometer head and the spectral distribution of the source to be measured should be known. It is important to note that there are photometer heads readily available that have an f_1' value of spectral responsivity less than 1.5%.

For determining total luminous flux of a luminaire or lamp using a goniophotometer, the photometer head shall have good cosine response within the angle range where light is incident, with the $f_2(\epsilon, \theta)$ value (relative deviation from the cosine function) of less than 2% within the acceptance angle range. In this application, the acceptance angle range is defined as the span of angles across which the photodetector will be expected to measure during a goniophotometric exploration. In most cases, this angle range will be less than 20 degrees. This good cosine response is required in order to accurately measure the candela values needed to calculate total luminous flux.

6.1.1.2 Angular Positioning. The uncertainty in angular position due to all factors (e.g., encoder accuracy, structural deflection, mirror alignment, reversal error) should be no more than 0.1 times the required measurement increment.

6.1.1.3 Repeatability of Light Readings. One purpose of goniophotometry is to accurately measure the spatial distribution of light for a lamp or luminaire. The goniophotometer should have good repeatability in light readings.^{3,4} There are many factors that contribute to the

uncertainty in light readings, and the goniophotometer should be evaluated to determine whether the system's measurement uncertainty is consistent with the measurement needs of the lamp or luminaire under test. Some factors influencing the system uncertainty are:

- Positioning repeatability
- Accuracy of gain ratios in amplifier and scaling circuits
- Positioning accuracy
- Quantization error in analog-to-digital (A/D) conversion
- Linearity in the A/D conversion
- Spectral responsivity of the photodetector vs. the source spectral properties

The uncertainty factors for any given goniophotometer depend on the design and construction of the instrument and the relative contributions of the factors appropriate for that system. As a general goal, light readings within a single test setup should have a repeatability of within $\pm 2\%$.

6.2 Test Distance

The test distance (see **Section 3.3**) shall be great enough so that the inverse-square law applies.¹ A test distance greater than or equal to 5 times the largest dimension of the luminous opening(s) should be sufficient for most roadway and area lighting luminaires. The luminaire optical properties should always be evaluated in regards to the minimum required distance. Test distances that are insufficient to satisfy the inverse-square law will cause errors in the reported luminous intensity values. In all cases, the test distance shall be reported. (See **Figure 6-1**.)

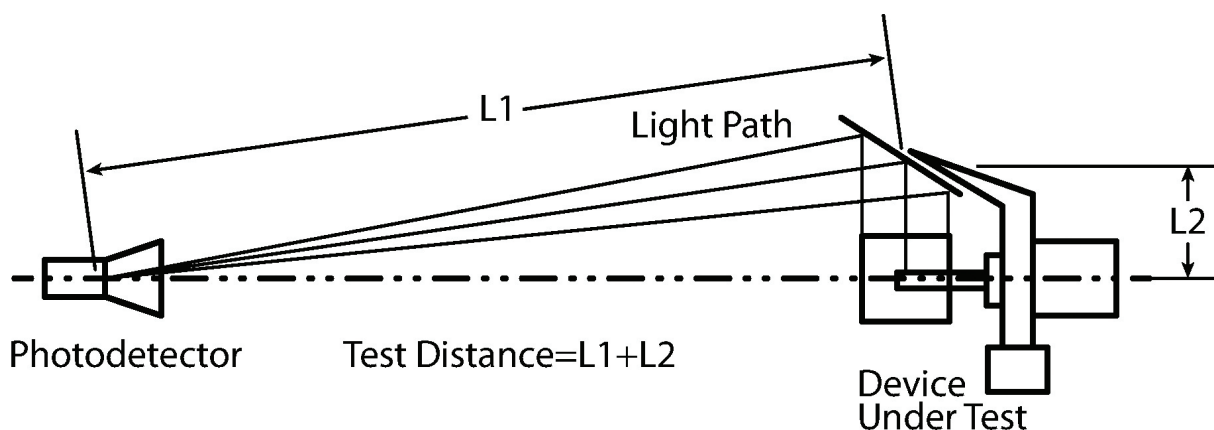


Figure 6-1. Diagram of test distance for a typical moving-mirror goniophotometer. (© Illuminating Engineering Society)

6.3 Selection and Preparation of Luminaire

6.3.1 Luminaire, Lamp, and Ballast Selection. The luminaire and lamp shall be selected to meet the requirements of the test(s). For HID lamps, ballasts regularly furnished as part of or recommended for use with the luminaire may be used to operate the lamp(s) during the test. As an alternative, ballasts that are compliant with American National Standards Institute (ANSI) definition of a “reference ballast” may be used. Due to the size of the reference ballasts, testing is performed with the reference ballast external to the fixture.

6.3.2 Cleaning of Luminaire Optical Parts. All optical parts shall be cleaned before measurements are made, unless the purpose of the test is to determine the effect of dirt on the luminaire. For the test that follows a dirt depreciation test, the luminaire shall be returned to as near “like new” condition as possible. Use of corrosive or abrasive materials in cleaning could alter the finish of the lamp, reflector, refractor or glass cover, and shall not be used.

6.4 Test Lamps

6.4.1 General. Lamps with stable light output are needed for photometric test purposes, and their light output should remain stable during a test or series of tests when constant line voltage is supplied. Test lamps should be selected for low eccentricity, low angular skew, and close conformance to the lamp manufacturer’s design dimensions and construction (specifically conformance to nominal light center length (LCL) dimensions).

6.4.2 Lamp Seasoning. Test lamps shall be seasoned (see **Section 3.2**) such that their characteristics remain constant during the test to be conducted. (Refer to *ANSI/IES LM-54-20, IES Guide to Lamp Seasoning* [see **Section 2.5**]), for detailed information.)

High intensity discharge (HID) lamps require special precautions when operating position is changed. Long stabilization periods (up to several days) may be necessary to re-stabilize the lamps in the new position. Even designated “universal operating” HID lamps

require prolonged stabilization in their actual operating position before being used as test lamps. In practice, separate lamps should be maintained for different operating positions (base up, base down, or horizontal) and should be operated only in their assigned position. Lamp seasoning shall also be done in the same position as that intended during testing.

6.4.3 Test Lamp Operation.

6.4.3.1 General. During photometric testing, stable lamp performance is of primary importance. As a result of their very different methods of operation, the stability of incandescent and HID lamps is determined in very different ways.

6.4.3.2 Incandescent Filament Lamps. Tungsten-halogen lamps shall be operated at a constant value of current near their nominal voltage. Other incandescent filament lamps should be operated at a constant current near 90% of rated input lamp voltage (at approximately 75% of nominal luminous flux this practice extends the life of test lamps). Current for incandescent lamps shall be regulated within $\pm 0.1\%$ of target current (DC operation may be necessary to achieve this current regulation requirement).

Operation at other voltages shall be specifically noted in the test report. When conducting absolute photometry, lamps should be operated under the conditions used to establish the nominal lumen rating.

For more information on electrical measurements, the reader should consult *ANSI/IES LM-45-20/R23, Approved Method: Electrical and Photometric Measurements of General Service Incandescent Filament Lamps* (see **Section 2.2**).

Before any measurements are taken, the seasoned lamps shall be operated long enough to reach stabilization. A stabilization time of ten minutes is usually sufficient for incandescent and tungsten-halogen lamps. A period of one minute of continuous operation, or up to three minutes for tungsten-halogen types, is usually sufficient for pre-burning. When the light output, and/or lamp voltage or current become stable, the lamp is stable.

Approved Method: Photometric Testing of Roadway and Area Lighting Luminaires Using Incandescent Filament or High Intensity Discharge Lamps

The following is a generally accepted four-step method for determining whether an incandescent filament lamp is stable:

- **Step 1.** Take five measurements of the lamp light output at fifteen-second intervals (total time = 1 minute). This period is in addition to the recommended pre-burning time.
- **Step 2.** Determine the maximum measured value, the minimum measured value, and the average value of the five consecutive measurements. Calculate the percent stability by dividing the difference between the maximum and minimum values by the average value of the five consecutive measurements.
- **Step 3.** If the value calculated in Step 2 does not exceed ¼ percent, the lamp is considered stable.
- **Step 4.** If the value calculated in Step 2 exceeds ¼ percent, the lamps are considered unstable, and measurements in 15-second intervals have to be continued until the last five measurements meet the criteria in Step 3.

This method is strongly recommended. Variation in the acceptance criterion and the number and duration of the measurement interval provide different levels of lamp stability, either more or less severe. Should a different method or criteria be used, it shall be so noted in the test report.

6.4.3.3 HID lamps. A stabilization time may range from one to three hours for mercury, high pressure sodium, and metal halide lamps. HID lamps operating on magnetic ballasts shall be operated within $\pm 2.0\%$ of the rated power of the lamp and regulated to within $\pm 0.5\%$. HID lamps operating on electronic ballasts shall have the ballast input voltage set to within $\pm 0.5\%$ of the unit's rated primary voltage and regulated to within $\pm 0.2\%$. If the rated primary voltage is a range, the test voltage shall be recorded in the test report. Metal halide lamps should not be moved after turn-off until the salts within the arc tube have cooled and solidified. For most applications, 30 minutes of cooling is sufficient.

For more information on electrical measurements, the reader should consult *ANSI/IES LM-51-20/R23, Approved Method: Electrical and Photometric Measurements of High Intensity Discharge Lamps* (see **Section 2.3**)

Before measurements are taken, the lamps should be operated until the electrical characteristics and light output are stable. In most cases, ten to thirty minutes are sufficient. Metal halide lamps may require considerably longer periods of stabilization time.

However, it is always better to judge stability from periodic checks of light output, lamp voltage/current, or both, rather than elapsed time. When light output, and/or lamp voltage or current become stable, the lamp is stable.

The following is a generally accepted four-step method for determining whether an HID lamp is stable:

- **Step 1.** Take four measurements of the lamp light output at one-minute intervals (total time = three minutes). This period is in addition to the recommended pre-burning time.
- **Step 2.** Calculate the percent difference between the maximum measured value and the minimum measured value for the four consecutive measurements.
- **Step 3.** If the value calculated in Step 2 does not exceed 1% for a mercury-vapor lamp, the lamp is considered stable. If the value calculated in Step 2 does not exceed 2½% for a high pressure sodium lamp, the lamp is considered stable. If the value calculated in Step 2 does not exceed 2% for a metal halide lamp, the lamp is considered stable.
- **Step 4.** If the value calculated in Step 2 exceeds the criteria in Step 3, the lamp is considered unstable, and additional measurements in one-minute intervals have to be continued until the last four measurements meet the criteria in Step 3.

This method is strongly recommended. Variation in the acceptance criterion and the number and duration of the measurement interval provide different levels of lamp stability, either more or less severe. Should a different method or criteria be used, it shall be so noted in the test report.

6.5 Test Methodologies

6.5.1 General. The lamp or luminaire shall be operated under conditions as typical of intended end use as possible, or suitable corrections made. For example, if a

luminaire is to be mounted on a pole at a tilted angle, it shall be mounted on the goniophotometer at this same angle.

6.5.2 Bare Lamp Measurement. When measuring the test lamp, the following conditions should be observed:

1. The lamp(s) shall be mounted and operated in the same orientation as they would be if they were in the luminaire. The light center of the test lamp shall be mounted at the goniophotometer center.
2. The same instruments (i.e., the photodetector and electrical measuring instruments) and test parameters (i.e., the test distance, electrical current for incandescent and tungsten halogen lamps, and wattage for HID) shall be the same as used for the luminaire.
3. Lamps and ballasts shall reach stable operating conditions before photometric readings are taken. Electronic monitoring should be used to determine when electrical and light readings become stable.
4. Lamp lumen ratings should be taken from the lamp manufacturer's product listing for the particular lamp used in the test.

6.5.3 Lamp Photometric Calibration Process. There are two basic methods to generate goniophotometric test data: relative photometry and absolute photometry. Each method has advantages and disadvantages.

In relative photometry, the goniophotometer is used to measure the test lamp independent of the luminaire, to create a value typically referred to as the *bare lamp sum*. The bare lamp sum is a number proportional to the lumen output of the lamp. This bare lamp sum is then used to normalize the intensity data of the luminaire. The normalized luminaire data can then be rescaled by the manufacturer's initial rated lumen output for the test lamp to create report data, or "IES file" data, that are prorated for the lamp's rated lumen output. From a practical standpoint, the relative method is desirable because a traceable photometric standard is not necessary.

In absolute photometry, the goniophotometer's responsivity in candelas or counts is established using a mathematical model and system component data (e.g.,

mirror reflectivity, photodetector responsivity), or by measuring the system response to a traceable standard of luminous intensity or luminous flux. The calibrated goniophotometer can then be used to measure the properties of a light source or a luminaire in terms of absolute luminous intensity. Absolute photometry is advantageous or necessary when light source properties in the luminaire are strongly influenced by thermal effects such that the sources cannot be independently measured. The drawback of absolute photometry is that it requires the maintenance of a traceable standard of luminous intensity.

6.5.3.1 Relative Photometry Method. The relative method consists of the following steps:

1. Measure the test lamp (i.e., the "bare lamp").
 - Measure the test lamp's luminous intensity distribution in the same position, under the same electrical operating conditions, and on the same goniophotometer as will be used for the luminaire test.
 - Use a measurement angle set appropriate for the lamp position. (See **Sections 6.5.3.1.1** and **6.5.3.1.2** for appropriate angle sets.)
2. Measure the luminaire.
 - Measure the luminaire's luminous intensity distribution using the same test lamp, under the same electrical operating conditions, and on the same goniophotometer as was used for the bare lamp test.
 - Use a measurement angle set appropriate for the luminaire's expected photometric distribution type. (See **Section 6.5.4.5** for appropriate angle sets.)
3. Process the luminous intensity data.
 - Determine the ratio (R_G) of the bare lamp test signal gain to the luminaire test signal gain.
 - Determine the manufacturer's initial lamp lumen rating for the test lamp.
 - Compute the test lamp's bare lamp sum by summing the product of the measured intensity data and the zonal constants for the measurement angle set. (See **Annex A** for information on computing zonal constants.)
 - Scale the measured intensity data to create "corrected" luminous intensity values that are

prorated to the test lamp's rated initial lumen output:

$$I_c = \frac{I_m \times LRR}{BLS} \times R_G,$$

where:

I_c = corrected luminous intensity value of luminaire at a given angle

I_m = measured luminous intensity value of luminaire at a given angle

BLS = bare lamp sum

LRR = lamp lumen rating

R_G = ratio of measurement gains for the bare-lamp and luminaire tests

4. Compute the luminaire efficiency.

- Compute the luminaire sum by summing the product of the measured luminaire intensity data and the zonal constants for the luminaire's measurement angle set. (See **Annex A** for information on computing zonal constants.)
- Compute the efficiency:

$$E_L = \frac{L_S}{BLS} \times R_G,$$

where:

E_L = luminaire efficiency

L_S = luminaire sum

BLS = bare lamp sum

R_G = ratio of measurement gains for the bare lamp and luminaire tests

It is not necessary to measure the bare lamp before the luminaire.

6.5.3.1.1 Vertically Mounted Lamps. A vertically mounted (i.e., vertical base up or vertical base down) test lamp's luminous output shall be determined by measurements in at least 16 lateral half-planes spaced equally about the longitudinal lamp axis.

The vertical angle increments within the half-planes shall be sufficiently dense to represent the luminous intensity distribution so that each data point is a good approximation to the average luminous intensity within its zone. For 10-degree zones, vertical angle measurements shall be at 0, 5, 15, 25, 35, ... 165, 175, and

180 degrees. If necessary, finer angular increments may be used to properly represent the intensity distribution.

6.5.3.1.2 Horizontally Mounted Lamps. Horizontally mounted test lamps shall be calibrated in the end-use position. The lamp shall be positioned about its longitudinal axes in a manner identical to that used for the luminaire. A horizontally mounted test lamp's luminous output shall be determined by measurements in at least 18 lateral half-planes spaced equally about the longitudinal lamp axis. Care should be taken to avoid measuring in planes that may be shadowed by the goniophotometer support system or other structural items (e.g., the 0° or 180° plane would typically contain a support structure for the lamp socket).

The vertical angle increments within the half-planes shall be sufficiently dense to represent the luminous intensity distribution so that each data point is a good approximation to the average luminous intensity within its zone. For 10-degree zones, vertical angle measurements shall be at 0, 5, 15, 25, 35 ... 165, 175, and 180 degrees. If necessary, finer angular increments may be used to properly represent the intensity distribution.

6.5.3.2 Absolute Photometry Method. There are two methods for conducting absolute photometry of roadway and area lighting incandescent filament and HID luminaires. In the first method, the goniophotometer detector system is calibrated using a standard of luminous intensity or standard of luminous flux (standard lamp). The second method is based on the use of a calibrated photometer.

6.5.3.2.1 Method Using Luminous Intensity Standard. A standard of luminous intensity is a light source that has been calibrated by a laboratory capable of establishing an absolute value of luminous intensity. A calibrated luminous intensity standard is mounted in the goniophotometer such that measurements can be made in the specified orientation and direction for which the luminous intensity standard is calibrated. The calibrated luminous intensity standard is powered in accordance with the instructions provided by the calibration laboratory, and measurements are made using the goniophotometer photodetector along the normal (i.e., perpendicular) light path inclusive of any system mirrors or other optical elements along the light path.

It is recommended that three standards be used to reduce the calibration uncertainty. The ratio of the calibrated luminous intensity values to the measured values of the calibrated standards are averaged to establish a calibration factor for the goniophotometer.

An example calculation is provided using three luminous intensity standards:

<u>Luminous Intensity Standards</u>	<u>Photometer Readings</u>
Standard 1: 1,500 cd	Standard 1: 3,782 counts
Standard 2: 1,550 cd	Standard 2: 3,908 counts
Standard 3: 1,532 cd	Standard 3: 3,862 counts

Calibration Factor

Cal Factor 1 = $1,500/3,782 = 0.3966$

Cal Factor 2 = $1,550/3,908 = 0.3966$

Cal Factor 3 = $1,532/3,862 = 0.3966$

Cal Factor Avg. = 0.3966 cd/count (at measurement gain settings)

This calibration factor is good for the system gain setting at which the calibration measurements were made. For other system gain settings, the factor is scaled in proportion to the gain ratios. Goniophotometers require a wide dynamic range to handle the range of candela values generated from bare lamps to beam-forming luminaires. Programmable amplifiers and high-resolution analog-to-digital converters normally provide this dynamic range. To establish an absolute system sensitivity that is independent of the gain settings, knowledge of the electrical signal processing is applied to compute the candelas per ampere of photocurrent. **Figure 6-2** shows a block diagram of typical signal processing in a goniophotometer.

6.5.3.2.2 Method Using Luminous Flux Standard. A standard of luminous flux is a light source that has been calibrated by a laboratory capable of establishing an absolute value of luminous flux. A calibrated luminous flux standard is mounted in the goniophotometer such that measurements can be made in the specified orientation for which the luminous flux standard is

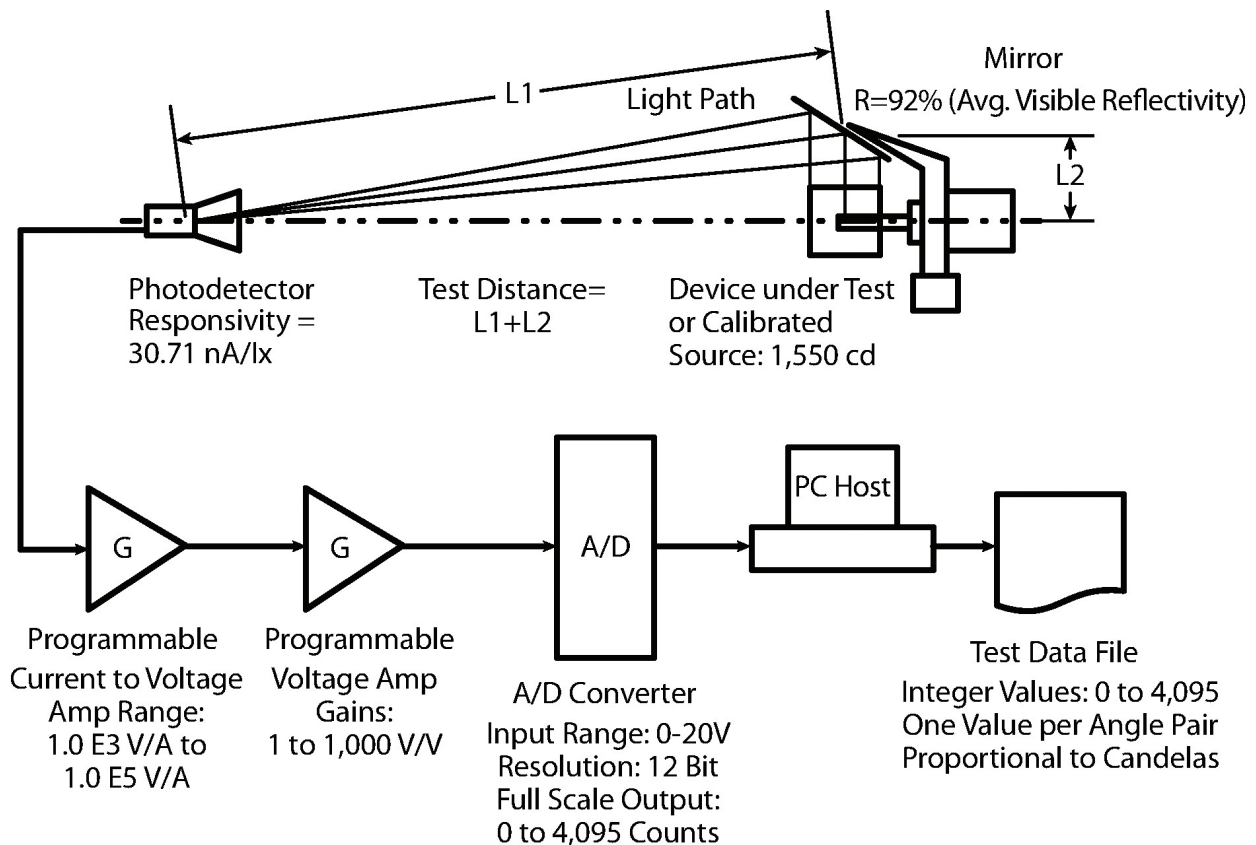


Figure 6-2. A block diagram of typical signal processing in a goniophotometer. (© Illuminating Engineering Society)

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calibrated (i.e., typically vertical base up or vertical base down). The calibrated luminous flux standard is powered in accordance with the instructions provided by the calibration laboratory, and measurements are made using the goniophotometer photodetector along the normal (i.e., perpendicular) light path inclusive of any system mirrors or other optical elements along the light path.

It is recommended that three standards be used to reduce the calibration uncertainty. The ratios of the calibrated luminous flux values to the measured values of the calibrated standards are averaged to establish a calibration factor for the goniophotometer.

An example calculation is provided using luminous flux standards:

<u>Luminous Flux Standards</u>	<u>Photometer Readings</u>
Standard 1: 5,420 lm	Standard 1: 13,670 counts
Standard 2: 5,456 lm	Standard 2: 13,756 counts
Standard 3: 5,382 lm	Standard 3: 13,572 counts

In this example, the counts are representative of the “lumen sum” of the luminous flux standard. In other words, the counts are representative of the sum of the products of the counts (proportional to the candelas) at each measurement angle and the zonal constant for the angular range of that measurement angle at the calibration gain setting.

Calibration Factor

$$\text{Cal Factor 1} = 5,420/13,670 = 0.3965$$

$$\text{Cal Factor 2} = 5,456/13,756 = 0.3966$$

$$\text{Cal Factor 3} = 5,382/13,572 = 0.3966$$

$$\text{Cal Factor Avg.: } 0.3966 \text{ lm}/(\text{count} \cdot \text{sr}) = 0.3966 \text{ cd}/\text{count} \\ (\text{at the calibration gain setting})$$

Note: $\text{cd} = \text{lm}/\text{sr}$

This calibration factor is good for the system gain setting at which the calibration measurements were made. For other system gain settings, the factor is

scaled in proportion to the gain ratios or calibrated directly at additional gain settings.

This system calibration factor (Sys Cal Factor) represents the system’s sensitivity inclusive of all optical factors such as path length, mirror reflectivity, and photodetector responsivity specific to the goniophotometer in use and can be scaled as necessary to account for different signal processing settings, such as amplifier gains.

6.5.3.2.3 Method Using Calibrated Photometer.

Absolute photometry using a calibrated photometer⁵ may be conducted by using the calibrated photometer as the detector system of the goniophotometer, or by transferring the calibration to the goniophotometer photodetector. The photometer’s calibration should be valid over the full range of measurement values encountered during testing of the bare lamp and the luminaire; therefore, aspects such as detector linearity and spectral mismatch should be considered and preferably corrected.

The goniophotometer system calibration is determined by using the calibrated photodetector and the reflectance of any system mirrors or transmission of other optical elements along the light path under the intended conditions of use. For example, if the goniophotometer being calibrated uses a back surface mirror with source and detector aligned at 42 degrees from the mirror surface normal, then the mirror reflectance, inclusive of the glass transmission losses, should be known at that angle of incidence. This system calibration is good for the determination of total luminous flux of a bare lamp or a luminaire. To determine the candela distribution of a source, the distance between the detector reference plane and the light source reference plane should be known. An example calculation is provided for both system calibrations (one for total luminous flux and one for luminous intensity).

Example of system calibration in lx/count using a calibrated photometer:

A national metrology institute (NMI)-traceable calibrated photometer corrected to the $V(\lambda)$ CIE photopic standard observer with f_1' of less than 3% has:

$$\text{Abs Cal} = 88.67 \text{ nA}/\text{lx} = 8.867 \times 10^{-8} \text{ A}/\text{lx}$$

$$\text{Abs Cal} = 1.128 \times 10^7 \text{ lx}/\text{A}$$

Gain Scaling Factor:

Preamp Transimpedance Gain (G1) = 1.0×10^5 V/A

Post-amp Gain (G2) = 100 V/V

Gain Factor = $G1 \times G2 = 1.0 \times 10^7$ V/A = 1.0×10^{-7} A/V

Analog-to-Digital Converter Parameters:

12 Bit A/D Converter: 4,096 counts, full scale

Input Signal Range: ± 10 V, full scale

A/D Factor = 4.8828125 mV/count

A/D Factor = 204.8 count/V

Mirror Reflectance (at angle of use):

Reflectance = 0.92 (for CIE illuminant A)

It is important that the mirror reflectance be representative of the spectral and geometric conditions of use in the goniophotometric measurements.

Calibration Factor (for this gain setting):

Cal Factor = Abs Cal $\times$ Gain Factor $\times$ A/D Factor/
Mirror Reflectance

Cal Factor = $(1.128 \times 10^7 \text{ lx/A})(1.0 \times 10^{-7} \text{ A/V})(4.883 \times 10^{-3} \text{ V/count}) / 0.92$

Cal Factor = $5.987 \times 10^{-3} \text{ lx/count}$

Example of system calibration in cd/count using a calibrated photometer:

Candela Calibration Factor (for this gain setting):

Cd Cal Factor = [Cal Factor (refer to previous example)][Distance (L1+L2)]²

Cd Cal Factor = $(5.987 \times 10^{-3} \text{ lx/count})(6.12\text{m} + 1.22\text{m})^2$

Cd Cal Factor = 0.3967 cd/count

For absolute photometry, the calibration of the standard lamp or the standard photometer shall be traceable to a national metrology institute (NMI). In addition, the laboratory should determine the measurement uncertainty when transferring standard values.

6.5.4 Luminaire Photometry. The following sections describe the physical setup for performing the photometric tests.⁶

6.5.4.1 Mounting. The luminaire should be mounted on a goniophotometer that is designed for minimum obstruction of light. Care should be exercised to prevent light from reflecting off the mounting apparatus directly

into the photodetector or back onto the luminaire where it may artificially increase luminaire intensities.

6.5.4.2 Luminaire Position Sensitivity. Luminaires can be sensitive to the effects of mounting orientation. Metal halide and incandescent lamps can exhibit differences in light output characteristics as a function of orientation.

Therefore, luminaires shall be tested only in their intended mounting orientation.

This precaution avoids test-related mechanical disturbances such as sag, warping, or shifts in the position of critical light controlling components; temperature variations due to changes in airflow; and changes in lamp intensity distribution due to orientation changes.

6.5.4.3 Luminaire Positioning on a Goniophotometer.

Luminaires shall be positioned correctly relative to the goniophotometer axes to correctly measure the luminous intensity distribution. Luminaires shall be positioned on the goniophotometer according to the following guidelines:

1. If the light center of the test lamp is above the reflector opening, or above the plane of the major light center of the reflector opening, the luminaire shall be mounted so that the plane of the major light-emitting opening coincides with the goniophotometer center. Examples of luminaires meeting this criterion and the location of the goniophotometer center are shown in **Figures 6-3** and **6-4**.
2. If the light center of the test lamp is clearly visible, and outside of any reflector, the luminaire shall be mounted on the goniophotometer so that the light center of the lamp is at the goniophotometer center. (See **Figure 6-5A**.)
3. If a translucent aperture obscures the light center of the test lamp, the luminaire shall be mounted so that the central plane of this aperture coincides with the goniophotometer center. (See **Figure 6-5B**.)

Note: For the above examples, if the luminaire uses more than one lamp, the geometric center of all lamp light centers is considered in place of the light center of any individual lamp.

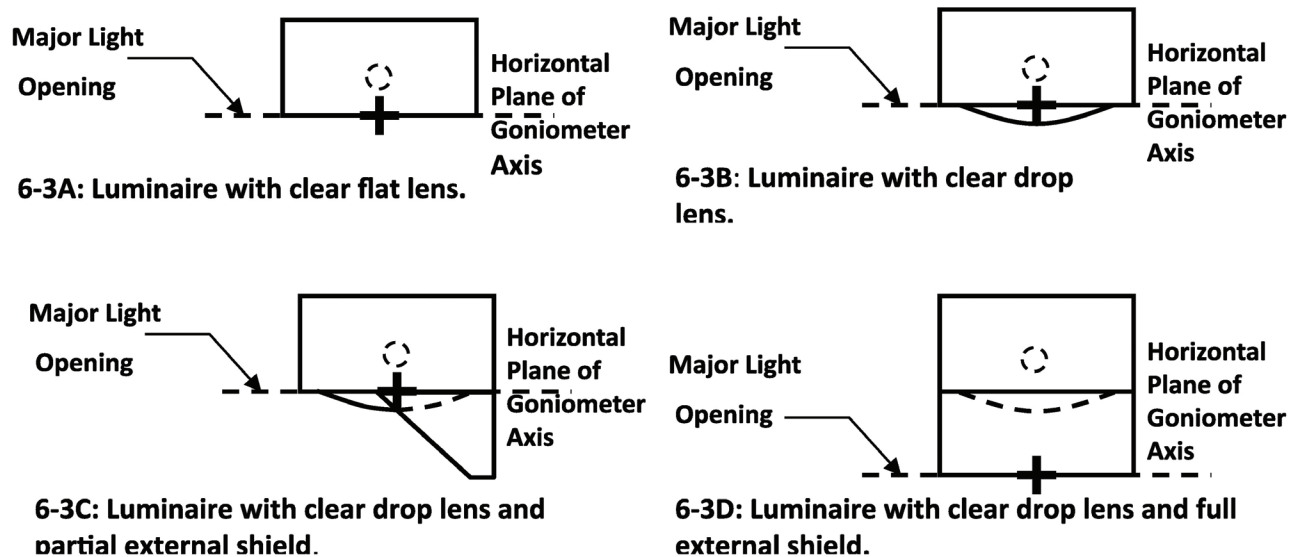


Figure 6-3. Examples of proper luminaire and goniophotometer positioning. The geometric light center of the lamp is above the major light opening. (© Illuminating Engineering Society)

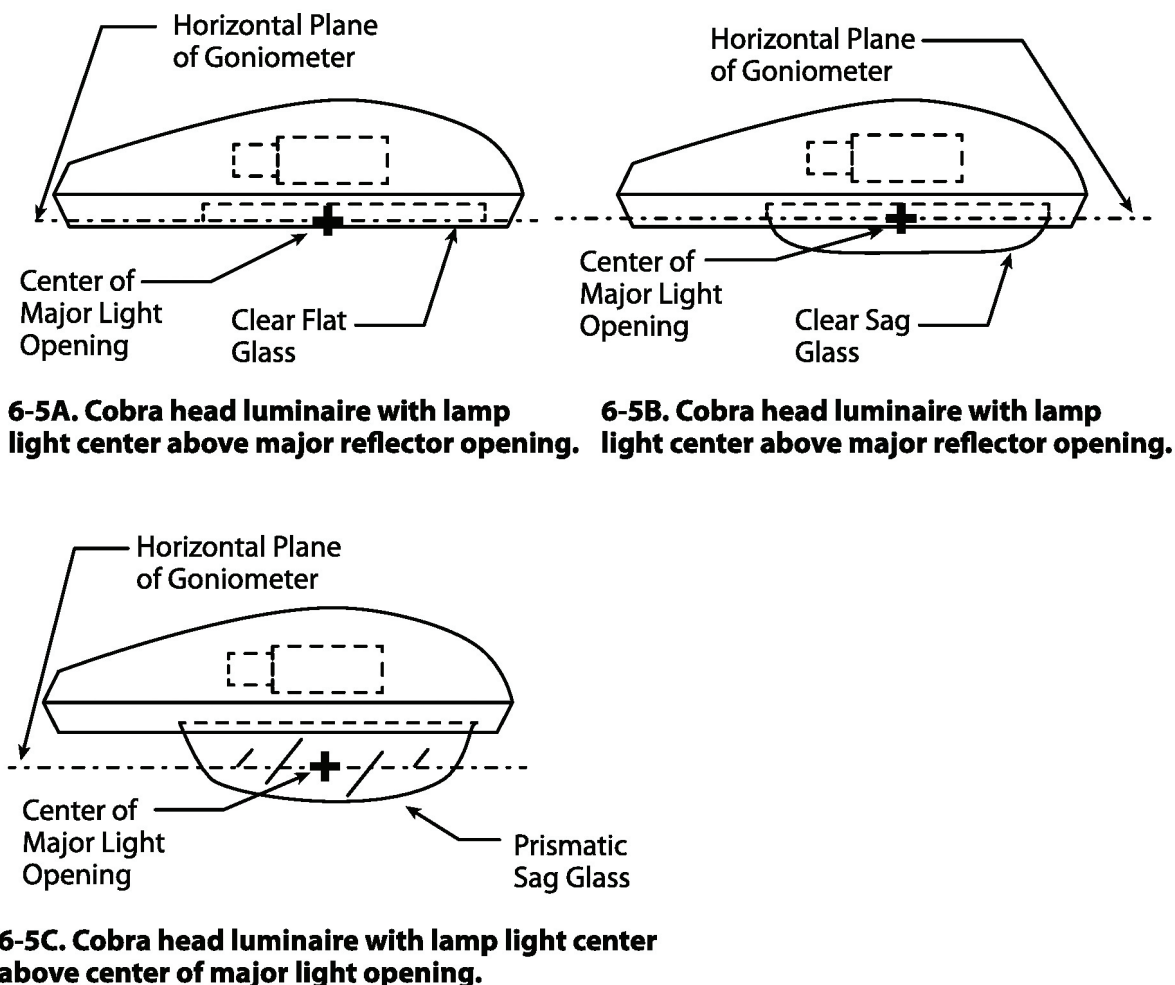
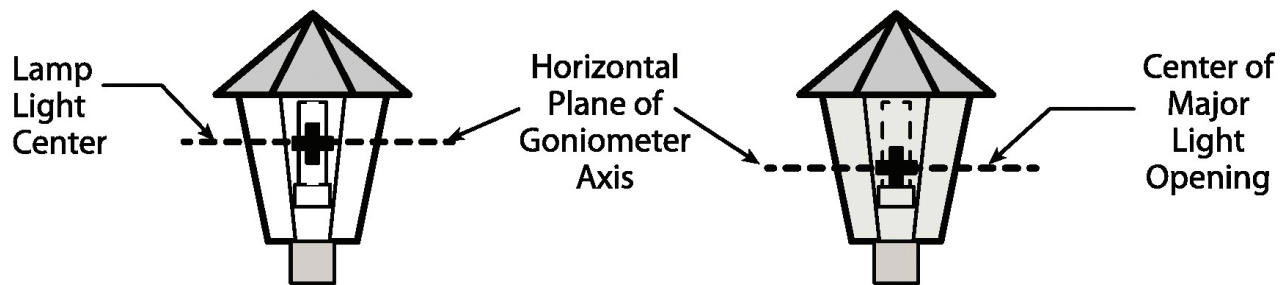


Figure 6-4. Cobra head luminaires illustrating various lamp light center cases. (© Illuminating Engineering Society)



6-5A. Luminaire with lamp light center clearly visible.

6-5B. Luminaire with lamp image obscured behind translucent aperture.

Figure 6-5. Examples of the proper luminaire and goniophotometer positioning. The light center is fully visible (6A) or partially obscured (6B). (© Illuminating Engineering Society)

6.5.4.4 Lamp Positioning

6.5.4.4.1 Luminaires with Incandescent Filament Lamps.

For lamps with a ring-type filament, the lamp shall be placed so that the open end of the filament is toward the street. [See **Figure 6-6(A)**.]

For lamps with a plane-type filament, the plane of the lamp filament shall be placed vertically. [See **Figure 6-6(B)**.]

6.5.4.4.2 Luminaires with HID Lamps. It is important that HID lamps are operated with the arc tube in the same position during the bare lamp calibration and the luminaire photometric test. It is suggested that the lamp base be marked so that the proper position can be verified any time the lamp is energized. Metal halide

lamps should not be moved after being turned off until the salts within the arc tube have cooled and solidified. For most applications, 30 minutes of cooling is sufficient.

For vertically mounted lamps, the lamp support rods shall lie in the 0- to 180-degree lateral plane. If there is only one support rod, it shall be located toward the house side (180 degrees). [See **Figure 6-7(A)**.]

For horizontally mounted lamps, the lamp support rods shall lie in the vertical plane. If there is only one support rod, it shall be located toward the zenith. For metal halide lamps that do not have an arc tube support rod, the exhaust tip shall be oriented toward zenith. It may be necessary to invert the socket to enable proper orientation of the arc tube support rod or exhaust tip. [See **Figure 6-7(B)**.]

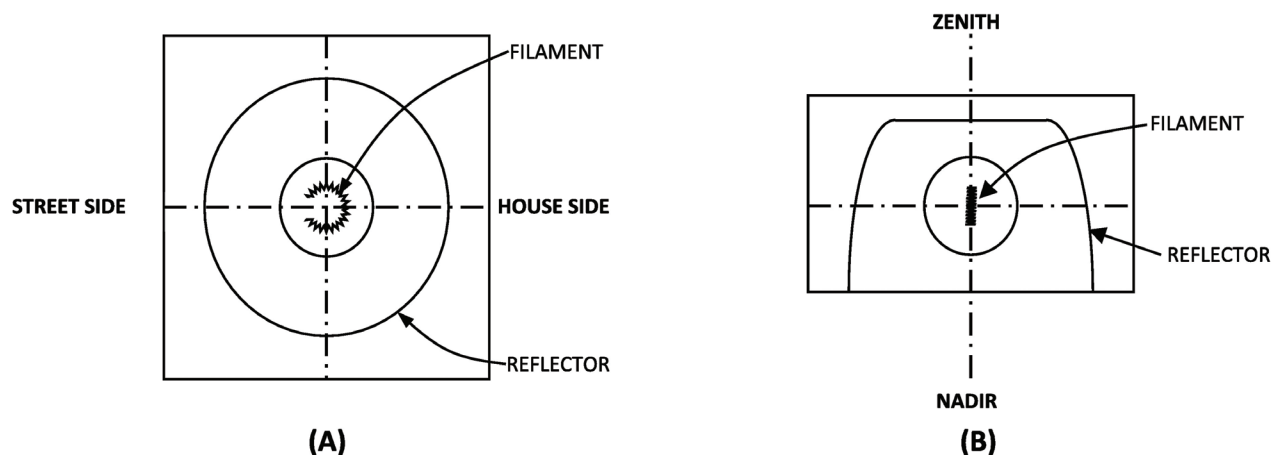


Figure 6-6. Lamp-reflector relationships: (A) vertically mounted ring-type filament incandescent lamps (plan view); (B) horizontally mounted plane-type filament incandescent lamps (side view). (© Illuminating Engineering Society)

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Using Incandescent Filament or High Intensity Discharge Lamps

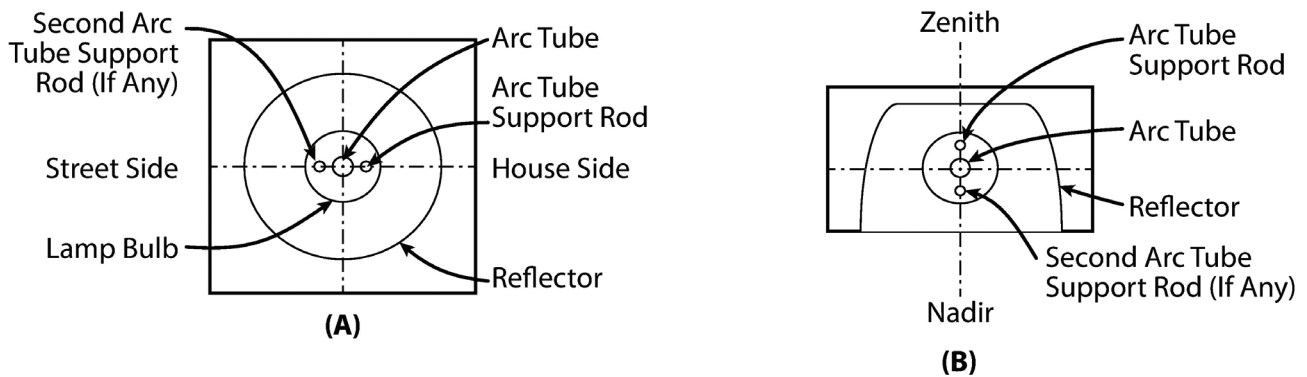


Figure 6-7. Lamp-reflector relationships: (A) vertically mounted HID lamps (plan view); (B) horizontally mounted HID lamps (side view). (© Illuminating Engineering Society)

Special consideration shall be given to the orientation of dual-arc high pressure sodium (HPS) lamps. There are two types of these lamps, one where both arcs operate at all times, and the other where only one arc operates at a time (a standby arc). For both types, if the luminaire design specifies an orientation for the arcs, the test shall be conducted with the lamps in this orientation. Otherwise, the arcs should be oriented in the same manner as the arc tube support rods would be oriented, with arc tube orientation superseding support rod positioning (**Figure 6-7**). In the case of a standby lamp, the primary arc should be treated the same as the first arc tube support rod. It may also be desirable to conduct a second test with the ignited arc in the secondary position. The test report shall clearly state the arc location used during the test.

6.5.4.5 Selection of Angles for Photometric Measurements

6.5.4.5.1 General. Angular relationships shall be as follows:

Vertical: Zero degrees at nadir, 90° at horizontal, and 180° at zenith.

Lateral: Measure counterclockwise, when viewing the luminaire from above, with 0° orthogonal to the curb toward the street side; 90° parallel to a curb line; 180° orthogonal to the curb toward the house side; and 270° parallel to a curb line.

Sufficient data shall be taken to accurately define

the luminaire's luminous intensity distribution. The number of horizontal half-planes and the number of vertical angles at which readings should be taken depends on the horizontal and vertical rates of change in the luminaire's luminous intensity distribution. Rapid changes in luminous intensity can occur near the peak of a distribution and can significantly affect calculated veiling luminance values. As readings approach the angle of maximum luminous intensity, the separation of readings may need to be reduced to ensure that the distribution is accurately represented. Readings should be taken in five-degree increments for vertical luminous intensity traces, but may need to be one degree or finer in regions with high rates of luminous intensity change.

Similarly, readings should be taken in no greater than 10-degree increments between horizontal half-planes, and may need to be reduced in the region of the horizontal maximum. Use of smaller reading increments should be considered when luminous intensity changes exceed 20% between readings.

During a typical goniophotometric scan, the nadir (0-degree vertical) and zenith (180-degree vertical) angles are typically measured multiple times (once for each measured lateral plane). These multiple measurements are repeat measurements of the same angular location, and therefore the nadir values should be averaged and reported for each plane. Similarly, the zenith values should be averaged and reported for each plane.

The point of maximum luminous intensity shall be located and reported.

6.5.4.5.2 Luminaires with IESNA Types I, I Four-Way, II, II Four-Way, III, IV, and VS Distribution. For luminaires having distributions that are symmetric about a single vertical plane, readings shall be taken in vertical planes that are a maximum of ten degrees apart. Additional planes of data may be necessary where the distribution is changing rapidly.

Averages may be made of the values taken at corresponding angles on opposite sides of the plane of symmetry, as shown in **Figures 6-8, 6-9, and 6-10**. Because of the averaging, it may be advantageous to laterally divide the distribution into ten-degree zones and measure at the mid-zone angles. Photometric

parameters, such as the location of the maximum candela value, may then be measured on one side of the plane of symmetry.

6.5.4.5.3 Luminaires with IES Type V Distribution. For luminaires having a distribution that is symmetric about the vertical axis, readings shall be taken in a minimum density of ten-degree lateral increments and then averaged.

6.5.4.5.4 Luminaires with Asymmetric Distribution. For luminaires having asymmetric distributions, readings shall be taken in lateral half-planes that are a maximum of ten degrees apart. It may be necessary to take additional planes of data in the region of the main beam or where the distribution is changing rapidly. Since there is no symmetry, any computations performed shall be computed without averaging.

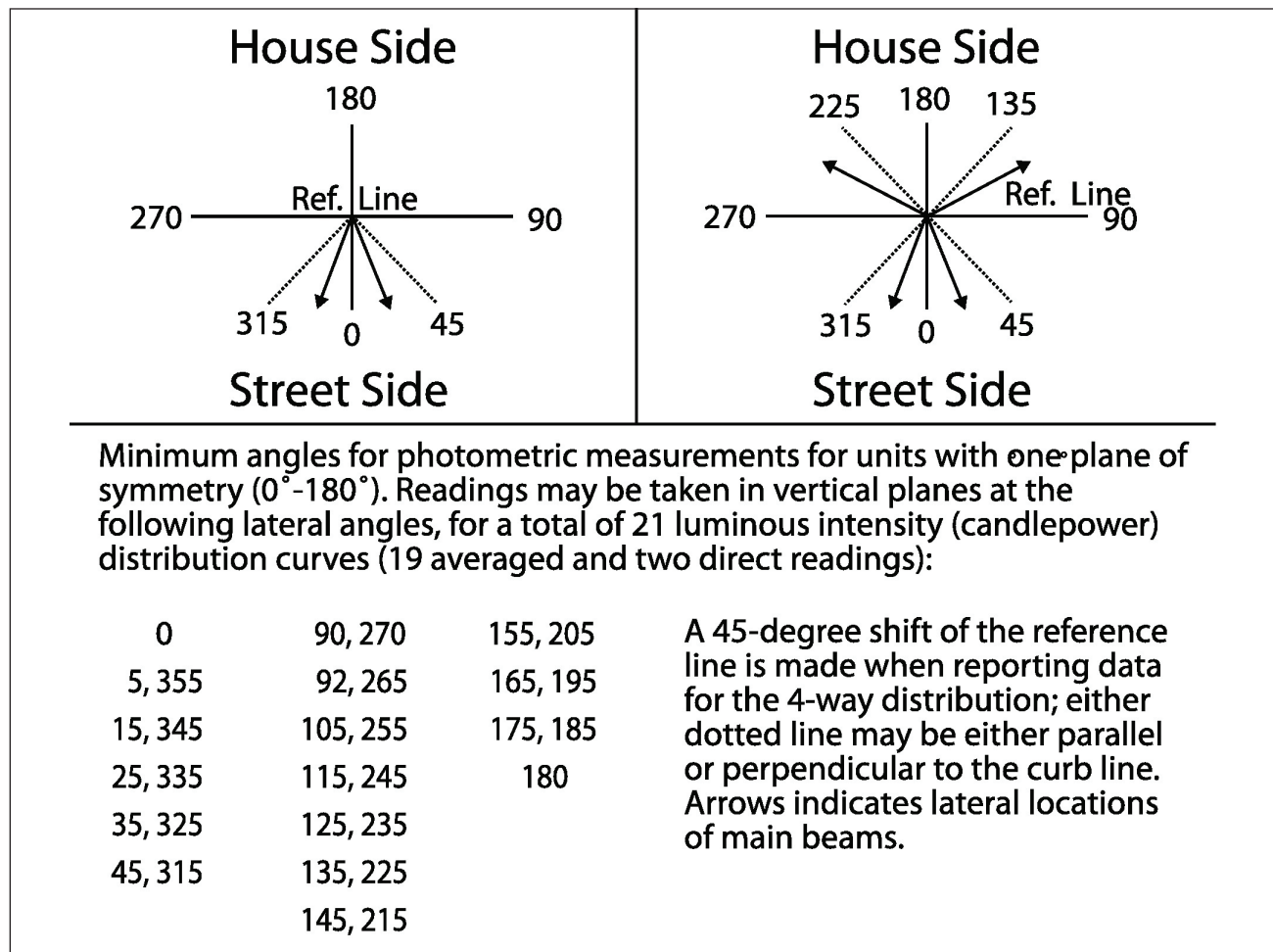


Figure 6-8. Averaging for distributions exhibiting bilateral symmetry (about the 0° – 180° lateral plane), such as Types II, III and IV. (© Illuminating Engineering Society)

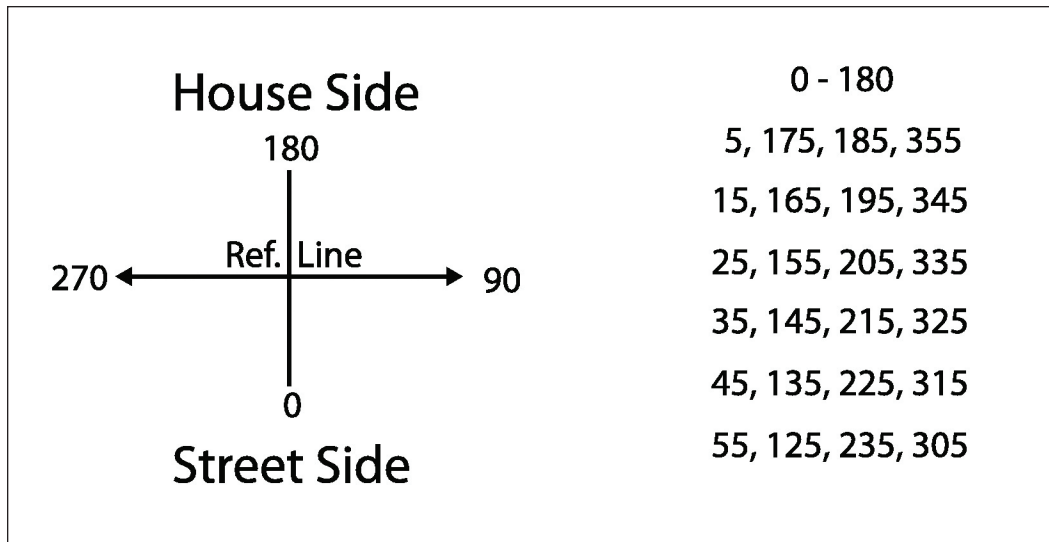


Figure 6-9. Averaging for distributions exhibiting four-quadrant symmetry (about the lateral planes 0 – 180 degrees and 90 – 270 degrees), such as Type I. (© Illuminating Engineering Society)

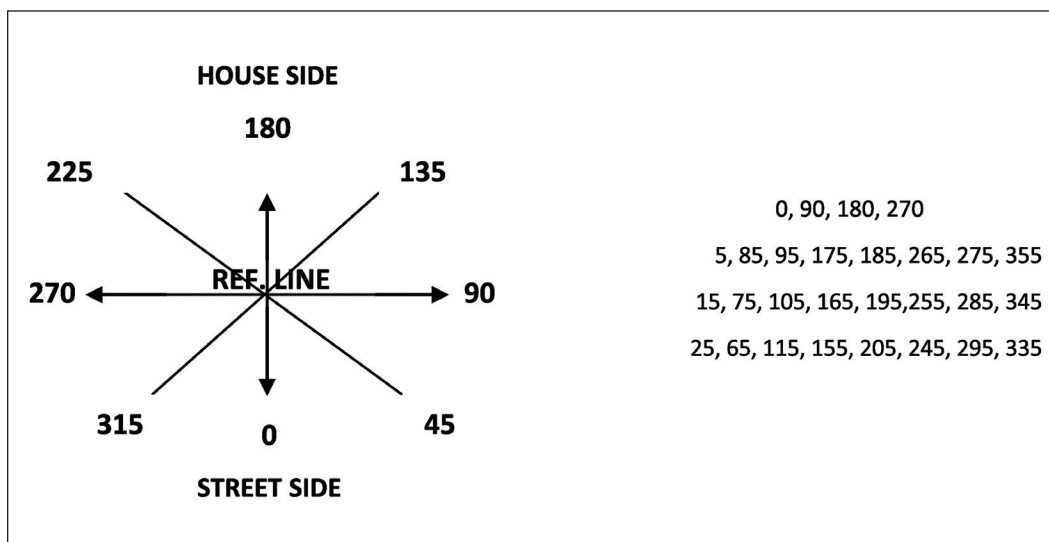


Figure 6-10. Averaging for distributions exhibiting octant symmetry (about the lateral planes 0 – 180, 45 – 225, 90 – 270, and 135 – 315 degrees), such as Type VS. (© Illuminating Engineering Society)

6.5.4.5.5 Area Luminaires. Some luminaires (such as outdoor wall packs, bollards, and general area lighting luminaires) may not be designed to fit into the standard Type I, II, III, IV, or VS classifications, and their light distribution may not be completely symmetric (as is the case in Type V luminaires, as described in **Section 6.5.4.5.3**). In this situation, the data shall be acquired in ten-degree or smaller lateral increments, and the luminaire should be classified as an area luminaire. If the light distribution of the luminaire exhibits bilateral, four-quadrant, or octant symmetry,

appropriate averaging as shown in **Figures 6-8, 6-9, and 6-10** may be applied.

6.5.5 Photometric Data Processing

6.5.5.1 Uplight Detection Threshold. Even with the most rigorous application of the techniques described in **Section 4.4** of this document, stray light will still contribute to the measured values in a photometric test. This can cause non-zero values to be measured at angles where the luminaire is not emitting light. These

non-zero values are particularly problematic when the test results are intended to verify compliance with specifications that require zero light at certain angles (notably, the BUG ratings of the Luminaire Classification System (LCS), described in ANSI/IES TM-15-20⁷).

For example, the luminaires depicted in **Figures 6-3A** and **6-4A** (see **Section 6.5.4.3**) are designed with a horizontal flat glass enclosure with no external light-directing devices. The design of this type of luminaire makes it impossible for light to be directed at or above 90 degrees vertical (90V) when the luminaire is mounted with the luminous opening horizontal. Because of the presence of stray light in the goniophotometer system, non-zero readings may still be recorded at these angles. If these spurious readings were reported as non-zero luminous intensity values, it would be impossible to classify the luminaire as “U0” according to TM-15-11.⁸

In order to avoid such misclassification of a luminaire, it is necessary to define the applicable test in terms of *detection*, rather than *measurement* of uplight. While it is impossible to verify that the uplight from any luminaire is exactly zero, it is possible to verify that a luminaire emits no uplight detectable by a particular measurement system. This necessarily involves determining the detection threshold of the measurement system and the comparison of the apparent uplight measured for the luminaire with that threshold.

For a luminaire designed to achieve zero uplight that is submitted for testing to support U0 classification, non-zero values that exist in the test data at and above 90V may be ignored or assigned a zero value only if analysis of the data collected at and above 90V is consistent with the design-based assumption that the device under test produces zero uplight. This is the case when the apparent uplight is below the detection threshold of the measurement system, and attributable to the effects of stray light and noise in the detector signal.

Determination of the uplight detection threshold of a particular measurement system involves consideration of various system characteristics, including stray light and photometric signal-to-noise performance. In order to ensure reproducibility of test results, a *conventional goniophotometer system* is defined as one that has

an uplight detection threshold equal to 0.25% of the total lumen output of the luminaire. Any measurement system used to perform U0 verification testing shall be shown to have an uplight detection threshold less than or equal to that of the conventional goniophotometer system. For the purpose of this standard, *detectable uplight* is defined as uplight detectable by the conventional goniophotometer system.

For a luminaire submitted for BUG testing, if the apparent uplight measured falls below 0.25% of the total lumen output of the luminaire, then luminous intensity values for angles at and above 90V may be omitted from the test report or assigned a zero value. The original data shall be retained for reference, and the details of the supporting detection threshold analysis shall be documented. In this case, the test report shall include the following statement:

“The device under test emits no detectable uplight, as defined by ANSI/IES LM-31-20/R23.”

If uplight values are assigned a zero value, the test report shall also include the statement:

“For the purpose of this report, certain non-zero uplight readings, attributable to instrument artifacts, have been assigned a zero value.”

6.5.5.2 Isoilluminance Diagram. An isoilluminance diagram of horizontal illuminance shall be shown on a horizontal plane. This plane shall have a width of at least 4 times the assumed mounting height (4 MH) on the street side of the reference line, 2 MH on the house side, and a length of 5 MH parallel to the reference line. (See **Annex B** for a discussion of the appropriate mathematical calculations.) It is recommended that enough isolines be shown so as to minimize the amount of interpolation needed. A numbering system such as 0.1 MH, 0.2 MH, 0.5 MH, 1.0 MH, 2.0 MH, 5.0 MH has been traditionally used, but a logarithmic system such as 0.1 MH, 0.2 MH, 0.4 MH, 0.7 MH, 1.0 MH, 2.0 MH, 7.0 MH often yields a more uniform pattern.

6.5.5.3 Isocandela Diagram. To classify the light distribution of the roadway luminaire in accordance with *ANSI/IES RP-8-18, Recommended Practice for Design and Maintenance of Roadway and Parking Facility Lighting*,⁸

at least a maximum candela point and a half-maximum candela trace are required. These intensity traces may be included as part of the isoilluminance diagram.

6.5.5.4 Utilization Efficiency. A pair of curves representing street- and house-side utilization of the emitted flux shall be plotted. These curves indicate the percentage of flux falling on the area (of any hypothetical width) in front of the luminaire (street side) and behind the luminaire (house side). The curves shall be marked as Street Side and House Side. Methods for computing utilization efficiency (luminous flux) may be found in **Annex C**.

6.5.5.5 Efficiencies. *Relative photometry:* To calculate the total efficiency of the luminaire, the lumens in all areas shall be summed as described in **Annex A** and divided by the lumen rating of the lamp(s).

Absolute photometry: To calculate the total efficacy of the luminaire, the lumens in all areas shall be summed as described in **Annex A** and divided by the electrical power (in watts) measured during the photometric test.

Light output shall be expressed as a percentage of lumen rating of the lamp (if appropriate) and as lumens subdivided into sections of a sphere as defined in the IES Luminaire Classification System (LCS)⁸ (refer to **Figure 6-11**):

- Uplight
- Forward Light
- Back Light

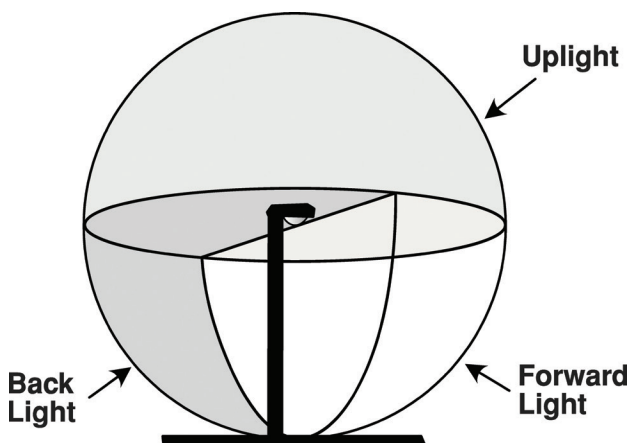


Figure 6-11. The three primary solid angles of the Luminaire Classification System (LCS).⁸ (© Illuminating Engineering Society)

7.0 Test Report

7.1 General

The test report shall describe all significant data concerning the luminaire tested, and the data should be listed so that the maximum amount of information can be interpreted.⁹

Section 7.2 identifies data items to be included in the test report. The sample report in **Annex D** illustrates typical data and information content for luminous intensity, lumen, and lux (footcandle) values, rather than actual test values.

7.2 Test Description

Test reports shall include all data and descriptive information necessary to adequately describe the lamp or luminaire under test and the test conditions. The goal of the data is to provide users of the report with sufficient information to verify that the contents of the report are relevant to the planned use of both the report and the associated "IES file" data. While the exact information recorded is specific to the devices being tested, and the format is somewhat specific to the testing agency (except as specified in **Section 6**), the recorded data shall provide the following types of information:

- 1) Basic test information
 - a) Test number
 - b) Test date
 - c) Test distance (light path distance from goniophotometer center to photodetector)
 - d) Testing agency
- 2) Luminaire description
 - a) Manufacturer's name; type and catalog number of luminaire, or indication of a prototype of test assembly
 - b) Reflector description (e.g., material [surface finish, texture, and specularly], reflectance [of painted surfaces], manufacturer's designation, dimensions)
 - c) Enclosure or refractor description (e.g., material[s], manufacturer's designation, dimensions)
 - d) Description of any auxiliary reflecting devices (e.g., louvers, shields)
 - e) Ballast description (e.g., catalog number[s], number of lamps, power factor correction, voltage, frequency, lamp current if measurable, ballast location)

- f) Luminaire mounting and orientation (e.g., horizontal or tilted, angle of tilt)
- 3) Test lamp description
 - a) Lamp catalog number, description, and number of lamps
 - b) Electrical parameters (e.g., lamp watts, input watts, input voltage [where applicable])
 - c) Rated lumen output at 25 °C (77 °F)
 - d) Lamp orientation during test
- 4) Results—for roadway and area lighting
 - a) IES photometric data file
 - b) Lumen output and efficiency of the luminaire, and total lumens on the house and the street side

The recorded data should also provide the following optional types of information:

- 5) Luminaire description
 - a) Graphic illustrations (e.g., housing and component shapes, dimensions, light center positions, goniophotometer center with respect to luminaire)
- 6) Results—For roadway and area lighting
 - a) Luminous intensity distribution curves in the vertical plane and in the cone containing the maximum luminous intensity value
 - b) Luminous intensity tabulation
 - c) Horizontal lux (footcandle) lines of equal illuminance (isoilluminance in lux or footcandles) computed for an actual mounting height (if unknown, 9 m [or 30 ft]) is recommended]. **Annex E** shows the conversion of luminous intensity to horizontal illuminance values. This may be a tabulation of values in lieu of a diagram.
 - d) Street-side and House-side Coefficient of Utilization (tabular or graphical)
 - e) Zonal lumen sums and efficiencies of the lighting distribution in accordance with IES TM-15-11⁸
 - f) Classification of the lighting distribution in accordance IES Area and Roadway “transverse” (i.e., Types I, II, III, IV, V, and VS) and “longitudinal” classifications (i.e., Short, Medium, and Long)⁹

- 7) Miscellaneous other information
 - a) Other useful information (e.g., a photograph)
 - b) Method(s) of luminaire sample selection
 - c) Arc voltage rise

Annex A – Zonal Constants for Conversion

The zonal constant is a factor by which the mean intensity emitted by a source of light in a given angular zone is multiplied to obtain the lumens in the zone.¹¹⁻¹⁵ These constants may also be used to calculate the lumens in any sector and thus the total lumens from a luminaire. Care should be taken to choose areas small enough to approach integration.

By multiplying the luminous intensity values found in each plane of measurement by the appropriate zonal constant, lumens for the corresponding segment of each zone are obtained. Addition of all lumen values corresponding to a given vertical angle of measurement will supply the lumens in the total zone represented.

For computer calculation, the zonal constants can be expressed by:

$$K = \frac{\pi(\cos\theta_1 - \cos\theta_2)(\phi_1 - \phi_2)}{180},$$

where:

θ_1 and θ_2 are vertical angles (in degrees) bounding the angular zone for a Type C coordinate system

ϕ_1 and ϕ_2 are lateral angles (in degrees) bounding the angular zone²

A simplified calculation may be made for applications where only the total lumens for a given vertical zone are of interest, such as when calculating bare lamp output. For each vertical angle, a weighted average of the test lamp's relative luminous intensity for all lateral angles shall be determined. The zonal constants when calculated for a total vertical zone can be calculated with the simplified equation:

$$K = 2\pi(\cos\theta_1 - \cos\theta_2)$$

Annex B – Procedure for Computing Iso-illuminance Curves

The procedure for computing isoilluminance curves is as follows:

- 1) Calculate horizontal illuminance values for a series of vertical angles for each lateral angle through which a vertical luminous intensity distribution curve has been taken. Vertical angles are typically taken in 5-degree increments from nadir to 60 degrees, in 2½-degree increments from 60 to 80 degrees, and in 1-degree increments from 80 to 86 degrees. Horizontal illuminance produced by light emitted at angles above 86 degrees is generally insignificant. Horizontal illuminance can be expressed by the formula:

$$E = \frac{I_{\theta}(\cos^3\theta)}{MH^2},$$

where:

I_{θ} = luminous intensity at a vertical angle of θ degrees

MH = mounting height of the luminaire

This is commonly referred to as the cosine-cubed law.

- 2) Plot horizontal illuminance values versus the tangent of the vertical angle. This may be computed on semi-log paper, or on rectangular coordinate paper if the coordinate (horizontal illuminance values) scale is changed each time that value goes below one-tenth full-scale value.
- 3) The isoilluminance diagram has as its scale a ratio of distance to mounting height, which is the tangent of the vertical angle. Radial lines may be drawn on the chart representing the intersection of the vertical planes through which luminous intensity values are recorded. Then, from the plotted curves of horizontal illuminance values versus tangents, desired horizontal illuminance values may be selected and plotted, and then values of equal illuminance may be joined by a smooth curve to form the isoilluminance diagram.

There are several computer graphical programs available that will facilitate the construction of isoilluminance diagrams following the general methodology described above. The input for these programs is a complete luminous intensity distribution array provided in standard IES format.¹⁰

Annex C – Computing Utilization Efficiency

To compute utilization efficiency, it is necessary to determine the portion of the luminaire's lumen output that actually strikes the pavement. There are several methods for completing this calculation. Two methods are presented in this Annex.

C.1 Method 1A

The Zonal Utilization Factors (ZUF) found in **Table C1** and corresponding zonal lumens shall be multiplied and summed to compute Utilization Efficiency. Zonal lumens shall be calculated from the luminous intensity data at the center of each calculation zone as shown in **Annex A**.

$$\sum_{i=1}^n \sum_{j=1}^m ZUF(i,j) \times Lumens(i,j),$$

where:

i = current horizontal zone (such as 0° to 10°, centered around the luminous intensity value at 5° horizontal)

j = current vertical zone (such as 30° to 40°, centered around the luminous intensity value at 35° vertical)

n = the number of horizontal zones; for **Table C-1**, $n = 9$

m = the number of vertical zones; for **Table C-1**, $m = 9$

Table C-1 shows ZUF values for one quadrant. ZUF values for the other quadrants would be equivalent. Depending on the symmetry of the luminaire, zonal lumens may vary between quadrants.

Table C-1. Zonal Utilization Factors

Angular Zones		Horizontal Angle†								
	Vertical Angle†	5	15	25	35	45	55	65	75	85
Ratio 0.5*	5	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	0.6	0.7	0.9	1	1	1	1	1	1
	35	0	0	0	0.1	0.5	0.9	1	1	1
	45	0	0	0	0	0	0.1	0.8	1	1
	55	0	0	0	0	0	0	0.1	0.9	1
	65	0	0	0	0	0	0	0	0.4	1
	75	0	0	0	0	0	0	0	0	0.7
	85	0	0	0	0	0	0	0	0	0.3
Ratio 1.0*	5	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	1	1	1	1	1	1	1	1	1
	35	1	1	1	1	1	1	1	1	1
	45	0.5	0.6	0.7	1	1	1	1	1	1
	55	0	0	0	0.1	0.5	0.9	1	1	1
	65	0	0	0	0	0	0.1	0.6	1	1
	75	0	0	0	0	0	0	0	0.5	1
	85	0	0	0	0	0	0	0	0	0.5
Ratio 2.0*	5	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	1	1	1	1	1	1	1	1	1
	35	1	1	1	1	1	1	1	1	1
	45	1	1	1	1	1	1	1	1	1
	55	1	1	1	1	1	1	1	1	1
	65	0.3	0.4	0.6	0.8	1	1	1	1	1
	75	0	0	0	0	0.1	0.4	0.8		1
	85	0	0	0	0	0	0	0	0.3	0.7
Ratio 3.0*	5	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	1	1	1	1	1	1	1	1	1
	35	1	1	1	1	1	1	1	1	1
	45	1	1	1	1	1	1	1	1	1
	55	1	1	1	1	1	1	1	1	1
	65	1	1	1	1	1	1	1	1	1
	75	0.1	0.2	0.3	0.5	0.7	0.9	1	1	1
	85	0	0	0	0	0	0	0.2	0.5	0.8

Table C-1. Zonal Utilization Factors *(continued)*

Angular Zones		Horizontal Angle†								
	Vertical Angle†	5	15	25	35	45	55	65	75	85
Ratio 4.0*	5	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	1	1	1	1	1	1	1	1	1
	35	1	1	1	1	1	1	1	1	1
	45	1	1	1	1	1	1	1	1	1
	55	1	1	1	1	1	1	1	1	1
	65	1	1	1	1	1	1	1	1	1
	75	0.5	0.6	0.7	0.9	1	1	1	1	1
Ratio 5.0*	85	0	0	0	0	0	0.2	0.4	0.6	0.9
	5	1	1	1	1	1	1	1	1	1
	15	1	1	1	1	1	1	1	1	1
	25	1	1	1	1	1	1	1	1	1
	35	1	1	1	1	1	1	1	1	1
	45	1	1	1	1	1	1	1	1	1
	55	1	1	1	1	1	1	1	1	1
	65	1	1	1	1	1	1	1	1	1
	75	0.8	0.9	1	1	1	1	1	1	1
	85	0	0	0	0.1	0.2	0.3	0.5	0.7	0.9

Table notes:

* Lateral distance across the street as a ratio of the mounting height (MH).

† Angles shown are at the center of 10 degree zones.

C.2 Method 1B

Method 1B is the same as Method 1A, except that the zonal utilization factors are calculated, allowing the use of angular zones or mounting height ratios other than those shown in **Table C-1**.

An algorithm for computing roadway Zonal Utilization Factors (ZUFs) is provided below. For this method, the following terms will be used:

Ratio: The lateral distance (across the street) as a ratio of the mounting height. The ratio may be any non-negative number. The typical Utilization Efficiency chart uses ratios in the range of 0 to 5.

Angular Zone: The size of the zone for which the ZUF and lumens are being calculated. In **Table C-1**, the angular zones are 10° x 10°.

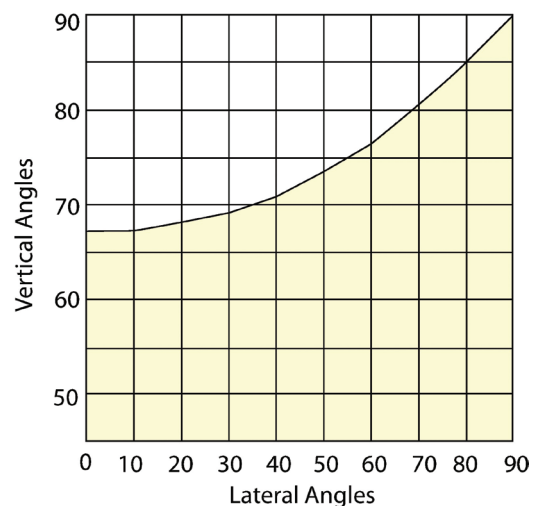


Figure C-1. Angular zones 10° x 10° for one quadrant. The shaded area illustrates the roadway passing through the angular zones. (© Illuminating Engineering Society)

For each ratio, it is necessary to calculate what portion of each angular zone falls on the roadway. In **Figure C-1**, for a ratio of 1.0, the shaded area would be on the roadway. Coverage is only shown for one quadrant, but would be the same for all four quadrants (two street-side and two house-side).

The Roadway ZUF is calculated from:

$$\text{Roadway ZUF} = \frac{A_{\text{polygon}}}{A_{\text{square}}}$$

where:

Roadway ZUF is the ZUF calculated for the roadway

A_{polygon} is the area of the shaded polygon

A_{square} is the area of the square

For each angular zone, the ZUF is between 0 (no roadway in the zone) and 1 (zone completely on the roadway).

Each angular zone is bounded by four angles: V_{HI} , V_{LO} , H_{HI} , H_{LO} , as shown in **Figure C-2**.

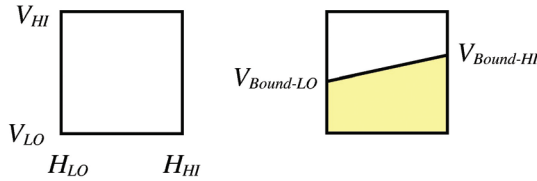


Figure C-2. Boundaries of one angular zone.

(© Illuminating Engineering Society)

For each of the two bounding horizontal angles (H_{HI} , H_{LO}), the vertical angle at which the edge of the roadway is intersected ($V_{\text{bound-HI}}$, $V_{\text{bound-LO}}$) is calculated:

$$V_{\text{bound}} = \tan^{-1}(\text{Ratio}/\cos H)$$

$$V_{\text{bound}} = \text{atan}(\text{Ratio}/\cos H)$$

If $V_{\text{bound-HI}} > V_{HI}$ and $V_{\text{bound-LO}} > V_{LO}$, then $ZUF = 1$

If $V_{\text{bound-HI}} < V_{LO}$ and $V_{\text{bound-LO}} < V_{LO}$, then $ZUF = 0$

For all other conditions, determine the portion of the angular zone that falls on the roadway.

Figure C-3 shows the four scenarios that can occur. The shaded area falls on the roadway.

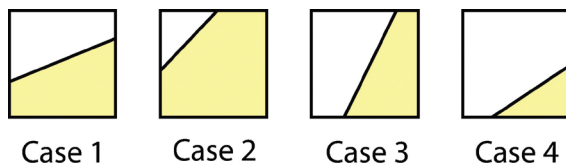


Figure C-3. Four possible cases for zones partially falling on the roadway. (© Illuminating Engineering Society)

Case 1:

$$A_{\text{polygon}} = \left(\frac{V_{\text{Bound-HI}} + V_{\text{Bound-LO}}}{2} - V_{LO} \right) \times (H_{HI} - H_{LO})$$

For Cases 2 through 4, calculation of H_{bound} values may be required in order to determine the coordinate of each polygon vertex:

$$H_{\text{Bound}} = \tan^{-1} \left[\frac{\sqrt{\tan^2 V^2 - \tan^2(\text{Ratio})^2}}{\text{Ratio}} \right]$$

Case 2:

$$A_{\text{polygon}} = A_{\text{square}} - \left(\frac{H_{\text{Bound-HI}} - H_{LO}}{2} \right) \times (V_{HI} - V_{\text{Bound-LO}})$$

Case 3:

$$A_{\text{polygon}} = \left(H_{HI} - \frac{H_{\text{Bound-HI}} + H_{\text{Bound-LO}}}{2} \right) (V_{HI} - V_{LO})$$

Case 4:

$$A_{\text{polygon}} = \left(\frac{H_{HI} - H_{\text{Bound-LO}}}{2} \right) (V_{\text{Bound-HI}} - V_{LO})$$

C.3 Method 2

A second method makes use of the isoilluminance diagram.

Step 1.

On the isoilluminance diagram, squares are laid off whose sides are 0.5 MH by 0.5 MH or smaller. At the center of these squares, the horizontal illuminance value is read. Lumens in any square may then be calculated for the fundamental equation:

$$\text{Lumens} = (\text{Average luminous intensity})(\text{Area}),$$

where:

Average luminous intensity is the horizontal value at the center of the Area

Area is the square of one-half the mounting height

Step 2.

Lumens in the square may then be summed in successive strips parallel to the curb line, each divided by the lumen rating used for the test lamp, and a plot of utilization versus transverse distance divided by mounting height may be made. It should be noted that if isoilluminance lines are shown for only one side of the axis of symmetry, the number of utilized lumen strips will double.

Annex D – Sample Test Report

This Annex shows an example of a photometric report for a high pressure sodium area luminaire, first in graphical format, then in text format.

TEST NO. : LM-31
CAT. NO. : LM31-COBRAHEAD

DATE: 3-21-2008

LUMINAIRE

SOCKET POS.: A, 2.0
REFLECTOR: ALUMINUM
SPECULAR
ENCLOSURE: CLEAR, FLAT GLASS

LAMP

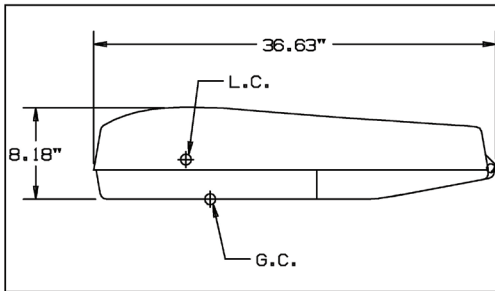
TYPE: 400 WATT HPS
ANSI: C400S51 I.D.: VTTI
ENVELOPE: ED-18 L.C.L.: 5.75 INCHES
1 LAMP(S) at LUMENS/LAMP: 50000

CLASSIFICATION

DISTRIBUTION: SHORT
TYPE: II
CONTROL: FULLCUTOFF

GENERAL

TEST DISTANCE: 36.3 FEET



	LUMENS	PERCENT OF LAMP
DOWNWARD STREETSIDE	26708	53.4
DOWNWARD HOUSESIDE	14280	28.6
UPWARD STREETSIDE	0	0.0
UPWARD HOUSESIDE	0	0.0
TOTAL	40988	82.0

To approximate performance for similar lamps with different Lumens, multiply Lumens, Lux and Footcandles by this ratio:

$$\text{RATIO} = \frac{\text{SELECTED LAMP LUMENS}}{50000}$$

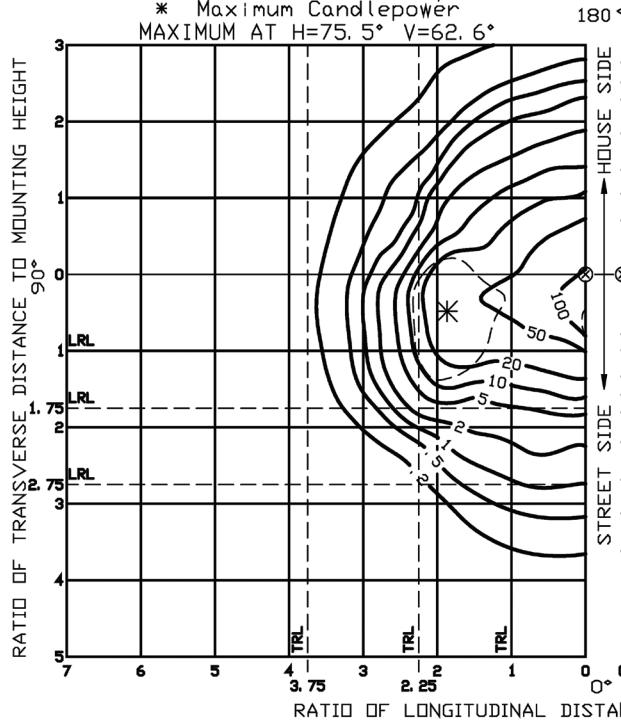
ISOLUX DIAGRAM

MOUNTING HEIGHT: 9.14 METERS

----- Half Maximum Candlepower Trace

* Maximum Candlepower

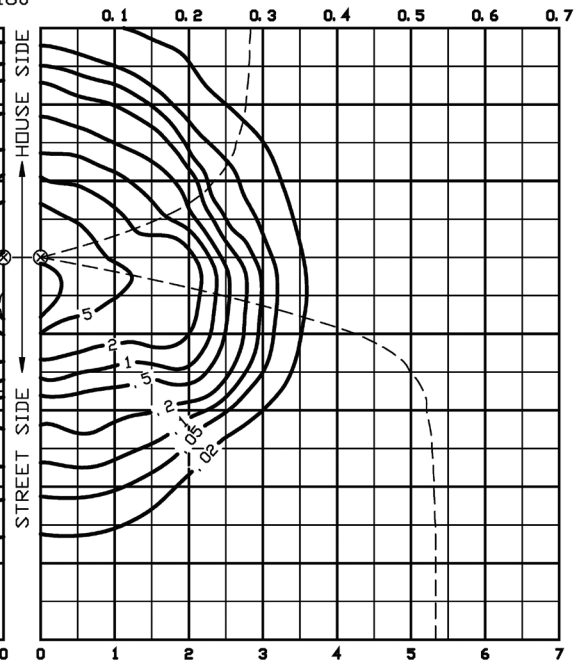
MAXIMUM AT H=75.5° V=62.6°



ISOFOOTCANDLE DIAGRAM

MOUNTING HEIGHT: 30 FEET

----- Coefficient of Utilization Curves



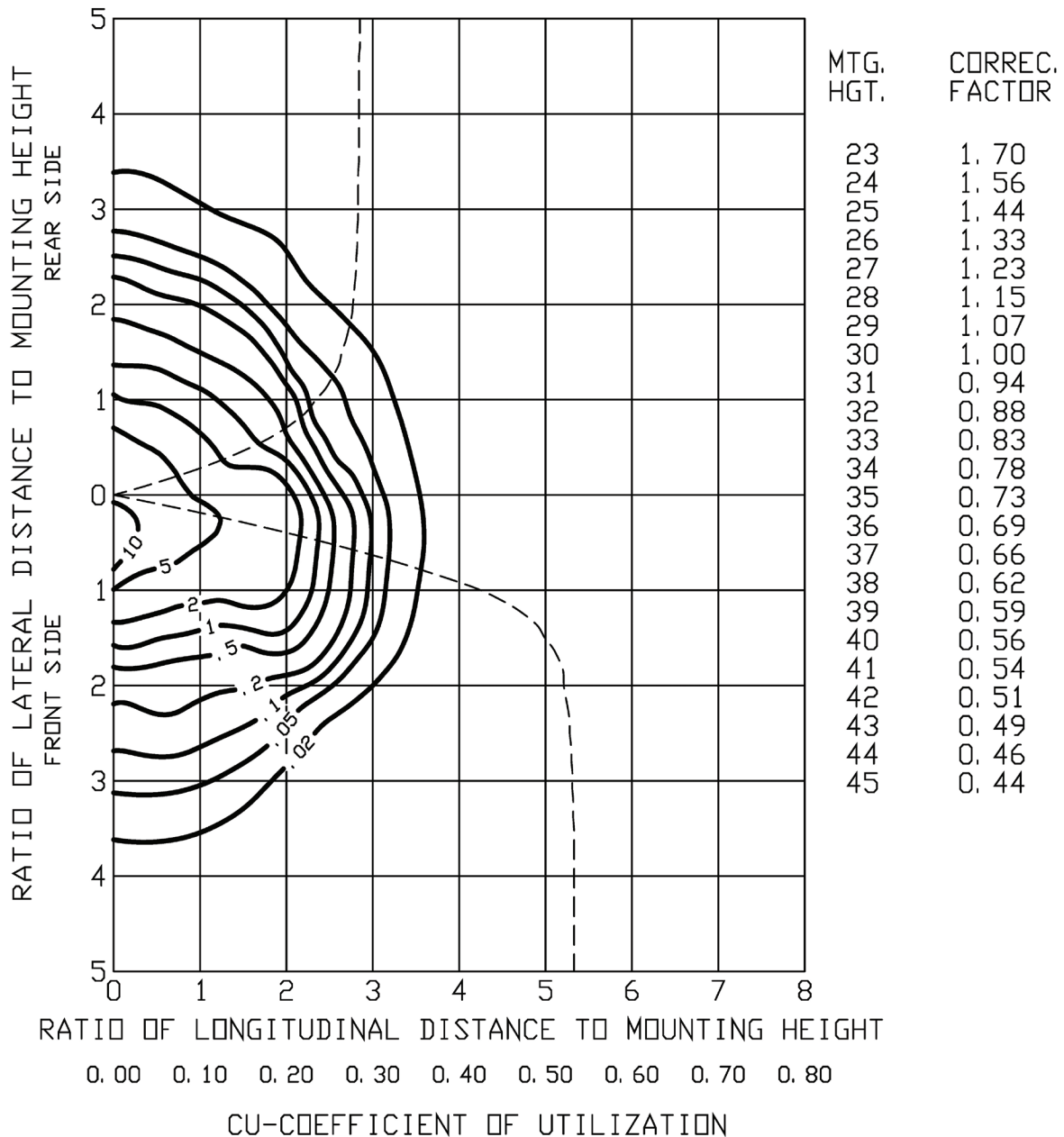
MOUNTING HEIGHT CORRECTION FACTORS

Mounting Height - Feet	18	22	26	30	34	38	42
Mounting Height - Meters	5.49	6.71	7.92	9.14	10.36	11.58	12.80
Factor	2.78	1.86	1.33	1.00	0.78	0.62	0.51

TESTED TO CURRENT IES AND NEMA STANDARDS UNDER STABILIZED LABORATORY CONDITIONS. VARIOUS OPERATING FACTORS CAN CAUSE DIFFERENCES BETWEEN LAB DATA AND ACTUAL FIELD MEASUREMENTS.

TEST NO. LM-31
 CAT. NO. LM31-COBRAHEAD
 DATE: 3-21-2008

ISOFOOT-CANDLE DIAGRAM
 VALUES BASED ON 30 FOOT MOUNTING HEIGHT



**Approved Method: Photometric Testing of Roadway and Area Lighting Luminaires
Using Incandescent Filament or High Intensity Discharge Lamps**

REPORT NO. LM-31

DATE 3-21-2008

CAT. NO. LM31-COBRAHEAD

TESTED WITH LAMP TYPE C400S51

LUMINAIRE CLASSIFICATION SYSTEM (LCS)

BUG Rating: B5 U0 G3

	Lumens	%Lamp
FL - Front-Low (0-30)	4562.4	9.1
FM - Front-Medium (30-60)	16115.0	32.2
FH - Front-High (60-80)	5985.7	12.0
FVH - Front-Very High (80-90)	45.2	0.1
BL - Back-Low (0-30)	3576.2	7.2
BM - Back-Medium (30-60)	8563.4	17.1
BH - Back-High (60-80)	2092.9	4.2
BVH - Back-Very High (80-90)	47.2	0.1
UL - Uplight-Low (90-100)	0.0	0.0
UH - Uplight-High (100-180)	0.0	0.0
Total LCS Lumens	40987.9	82.0

REPORT NO. LM-31

DATE 3-21-2008

CAT. NO. LM31-COBRAHEAD

TESTED WITH LAMP TYPE C400S51

CANDELA

NADIR C.P. = 8624., MAX. C.P. = 35518.

LATERAL ANGLE

	0.0	5.0	15.0	25.0	35.0	45.0	55.0	65.0	75.0	85.0
0.0	8624.	8624.	8624.	8624.	8624.	8624.	8624.	8624.	8624.	8624.
5.0	8783.	8813.	8782.	8689.	8606.	8521.	8471.	8434.	8426.	8413.
15.0	12061.	11839.	10794.	10413.	10624.	10106.	9590.	8978.	8611.	8441.
25.0	18129.	16261.	12968.	13985.	12228.	11562.	11155.	10341.	9427.	8835.
35.0	19905.	16760.	13227.	12154.	12808.	12191.	12746.	12339.	11488.	10249.
45.0	12357.	11821.	9716.	10606.	10820.	12857.	13745.	15928.	16427.	12875.
55.0	7352.	8099.	7113.	8175.	9201.	12339.	14725.	19110.	22458.	19535.
60.0	4514.	4978.	4915.	5624.	7302.	10502.	19665.	27564.	29913.	29562.
V 62.5	3160.	3298.	3755.	4360.	6112.	9501.	21681.	30228.	31985.	31042.
E 65.0	2619.	2708.	3217.	3548.	4673.	7572.	21015.	29303.	28470.	26028.
R 67.5	2346.	2425.	3136.	3015.	3657.	6471.	18074.	21163.	16982.	13301.
T 70.0	2061.	2172.	2603.	2564.	2967.	3903.	9225.	9472.	6756.	5183.
I 72.5	1532.	1624.	1769.	1739.	1669.	1392.	2473.	3119.	2286.	1857.
C 75.0	834.	903.	951.	789.	394.	482.	716.	954.	869.	820.
A 77.5	356.	335.	397.	241.	236.	282.	382.	455.	435.	443.
L 80.0	210.	192.	168.	168.	176.	200.	238.	264.	260.	268.
82.5	129.	115.	110.	115.	120.	134.	144.	159.	159.	159.
<u>A</u> 85.0	64.	67.	64.	71.	71.	80.	84.	90.	91.	91.
N 90.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
G 95.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
L105.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E115.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
125.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
135.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
145.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
155.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
165.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
175.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
180.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

**Approved Method: Photometric Testing of Roadway and Area Lighting Luminaires
Using Incandescent Filament or High Intensity Discharge Lamps**

REPORT NO. LM-31

DATE 3-21-2008

CAT. NO. LM31-COBRAHEAD

TESTED WITH LAMP TYPE C400S51

CANDELA

NADIR C.P. = 8624., MAX. C.P. = 35518.

LATERAL ANGLE

	90.0	95.0	105.0	115.0	125.0	135.0	145.0	155.0	165.0	175.0
0.0	8624.	8624.	8624.	8624.	8624.	8624.	8624.	8624.	8624.	8624.
5.0	8412.	8417.	8406.	8412.	8406.	8434.	8441.	8493.	8511.	8585.
15.0	8388.	8397.	8325.	8227.	8079.	8001.	7999.	8112.	8354.	8591.
25.0	8717.	8722.	8745.	8641.	8650.	8384.	8238.	8191.	8689.	9745.
35.0	9903.	9505.	9053.	8626.	7864.	7611.	6983.	6908.	7181.	7936.
45.0	11283.	10206.	8891.	7808.	6904.	6356.	5938.	5709.	5126.	5219.
55.0	16279.	12567.	8382.	7217.	6342.	5722.	4926.	4693.	4562.	4157.
60.0	25547.	19720.	9314.	6743.	5737.	4784.	4109.	3898.	4098.	4242.
V 62.5	26491.	19332.	7612.	6340.	5348.	4423.	3726.	3528.	3197.	3461.
E 65.0	19831.	13005.	5748.	5309.	4310.	3687.	3067.	2849.	2555.	3095.
R 67.5	8602.	4690.	2107.	1968.	1811.	2050.	1989.	1913.	1713.	1963.
T 70.0	3395.	2290.	1437.	1317.	1018.	886.	970.	988.	976.	1130.
I 72.5	1522.	1221.	992.	936.	747.	655.	766.	687.	728.	839.
C 75.0	735.	679.	615.	583.	489.	452.	533.	515.	672.	779.
A 77.5	426.	433.	389.	395.	325.	287.	288.	338.	542.	682.
L 80.0	258.	280.	245.	251.	206.	186.	170.	193.	351.	551.
82.5	152.	162.	147.	148.	130.	120.	106.	112.	139.	343.
<u>A</u> 85.0	88.	92.	85.	88.	77.	74.	62.	65.	59.	80.
N 90.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
G 95.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
L105.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E115.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
125.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
135.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
145.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
155.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
165.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
175.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
180.0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

1	REPORT NO. LM-31	DATE 3-21-2008
2	CAT. NO. LM31-COBRAHEAD	
3	TESTED WITH LAMP TYPE C400S51	
4		CANDELA
5		NADIR C.P. = 8624., MAX. C.P. = 35518.
6		LATERAL ANGLE
7		180.0
8	0.0	8624.
9	5.0	8646.
10	15.0	8691.
11	25.0	9978.
12	35.0	8110.
13	45.0	5816.
14	55.0	4262.
15	60.0	4366.
16	V 62.5	3774.
17	E 65.0	3445.
18	R 67.5	2120.
19	T 70.0	1177.
20	I 72.5	842.
21	C 75.0	763.
22	A 77.5	684.
23	L 80.0	588.
24	82.5	379.
25	<u>A</u> 85.0	77.
26	N 90.0	0.
27	G 95.0	0.
28	L 105.0	0.
29	E 115.0	0.
30	125.0	0.
31	135.0	0.
32	145.0	0.
33	155.0	0.
34	165.0	0.
35	175.0	0.
36	180.0	0.

**Approved Method: Photometric Testing of Roadway and Area Lighting Luminaires
Using Incandescent Filament or High Intensity Discharge Lamps**

38 REPORT NO. LM-31 DATE 3-21-2008
39 CAT. NO. LM31-COBRAHEAD
40 TESTED WITH LAMP TYPE C400S51
41

42 CONSTANTS

43 LAMP CONSTANT 147.993
44 NUMBER OF LAMPS 1
45 VENDOR LUMENS/LAMP 50000.
46 PER UNIT M.H. 30.0
47 TOTAL LUMENS 40988.
48 TOTAL EFFICIENCY 82.0
49

50 LUMENS BY VERTICAL QUADRANTS

	STREET SIDE		HOUSE SIDE	
	LUMENS	PCT	LUMENS	PCT
53 DOWN	26708.	53.4	14280.	28.6
54 UP	0.	0.0	0.	0.0

55

56 UTILIZATION TABLE

	STREET SIDE		HOUSE SIDE	
58 MOUNTING	UTILIZED		<u>UTILIZED</u>	
59 HEIGHT RATIO	LUMENS	EFF.	LUMENS	EFF.
60 0.000	0.0	0.0	0.0	0.0
61 0.500	12305.3	24.6	7951.7	15.9
62 1.000	21244.6	42.5	11699.3	23.4
63 1.500	24969.3	49.9	13206.1	26.4
64 2.000	26039.8	52.1	13831.5	27.7
65 2.500	26392.4	52.8	14054.0	28.1
66 3.000	26549.9	53.1	14132.9	28.3
67 4.000	26648.7	53.3	14201.4	28.4
68 5.000	26671.0	53.3	14231.8	28.5

69	(INTERPOLATED VALUES)				
70	0.250	6573.4	13.1	4501.4	9.0
71	0.750	17195.7	34.4	10351.0	20.7
72	1.250	23758.7	47.5	12732.8	25.5
73	1.750	25836.3	51.7	13629.0	27.3
74	2.250	26305.8	52.6	13993.1	28.0
75	2.750	26495.5	53.0	14111.4	28.2
76	3.250	26601.6	53.2	14161.2	28.3
77	3.500	26635.3	53.3	14182.1	28.4
78					
79	NOTES:				
80	-LM31 SAMPLE REPORT				
81	-SOCKET POSTION: A, 2.0				
82	-RESULTS OF THIS TEST WERE: SHORT, TYPE II, B5 U0 G3				

Annex E – Table of Constants for Horizontal Illuminance

This Annex provides pre-calculated constants that can be multiplied by luminaire luminous intensity (I) to obtain horizontal illuminance (E).

$$E = I \times C_E$$

where:

$C_E = \cos^3(\text{Vertical Angle})/(\text{Mounting Height})^2$ and can be obtained from **Table E-1**

E = Horizontal Illuminance at the target location

I = Luminous Intensity in the direction of the target

Vertical Angle is the vertical angular deflection to the target

Table E-1. Table of Constants for Horizontal Illuminance (C_E)

Vertical Angle	Mounting Height (Meters or Feet)			
	6	7.5	9	Tangent
0	0.02778	0.01778	0.01235	0.000
5	0.02746	0.01758	0.01221	0.087
10	0.02653	0.01698	0.01179	0.176
15	0.02503	0.01602	0.01113	0.268
20	0.02305	0.01475	0.01024	0.364
25	0.02068	0.01323	0.009191	0.466
30	0.01804	0.01155	0.008019	0.577
35	0.01527	0.009772	0.006786	0.700
40	0.01249	0.007992	0.005550	0.839
45	0.009821	0.006285	0.004365	1.000
50	0.007377	0.004721	0.003279	1.192
55	0.005242	0.003355	0.002330	1.428
60	0.003472	0.002222	0.001543	1.732
65	0.002097	0.001342	0.0009319	2.145
70	0.001111	0.0007113	0.0004939	2.747
75	0.0004816	0.0003082	0.0002140	3.732
78	0.0002497	0.0001598	0.0001110	4.705
80	0.0001454	0.00009309	0.00006464	5.671
81	0.0001063	0.00006806	0.00004726	6.314
82.5	0.00006177	0.00003953	0.00002745	7.596
83	0.00005028	0.00003218	0.00002235	8.144
84	0.00003172	0.00002030	0.00001410	9.514
85	0.00001839	0.00001177	0.000008173	11.430
86	0.000009429	0.000006034	0.000004191	14.301

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